

Introduction to single-cell methylation analysis

G&E Data Science Working Group

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Lecture overview

My main take-aways:

1. Most single cell omics analyses are essentially the same.
2. DNA methylation has unique attributes that allow for unique analyses.
3. I'm excited about DNA methylation and I think you should be too.

INTRO

- Unique samples for DNA methylation
- Foundations of methylation
- How we can measure it in single-cell scale

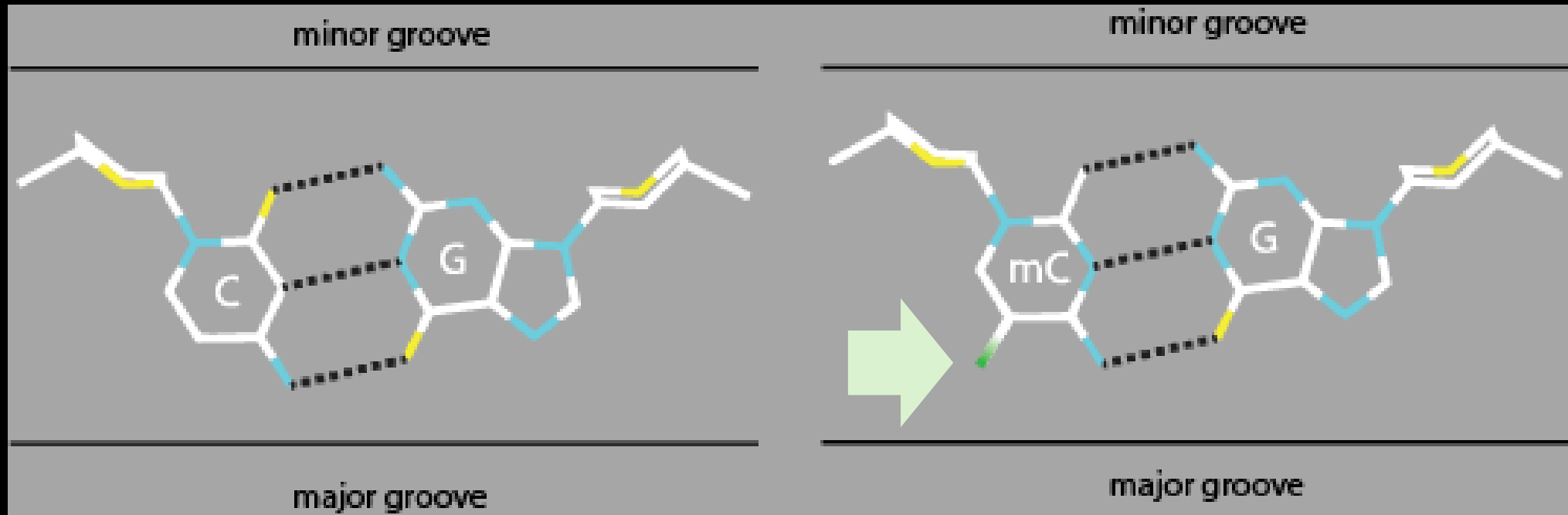
ANALYSIS

- Analysis of most-omics data sets
- How methylation analysis is different

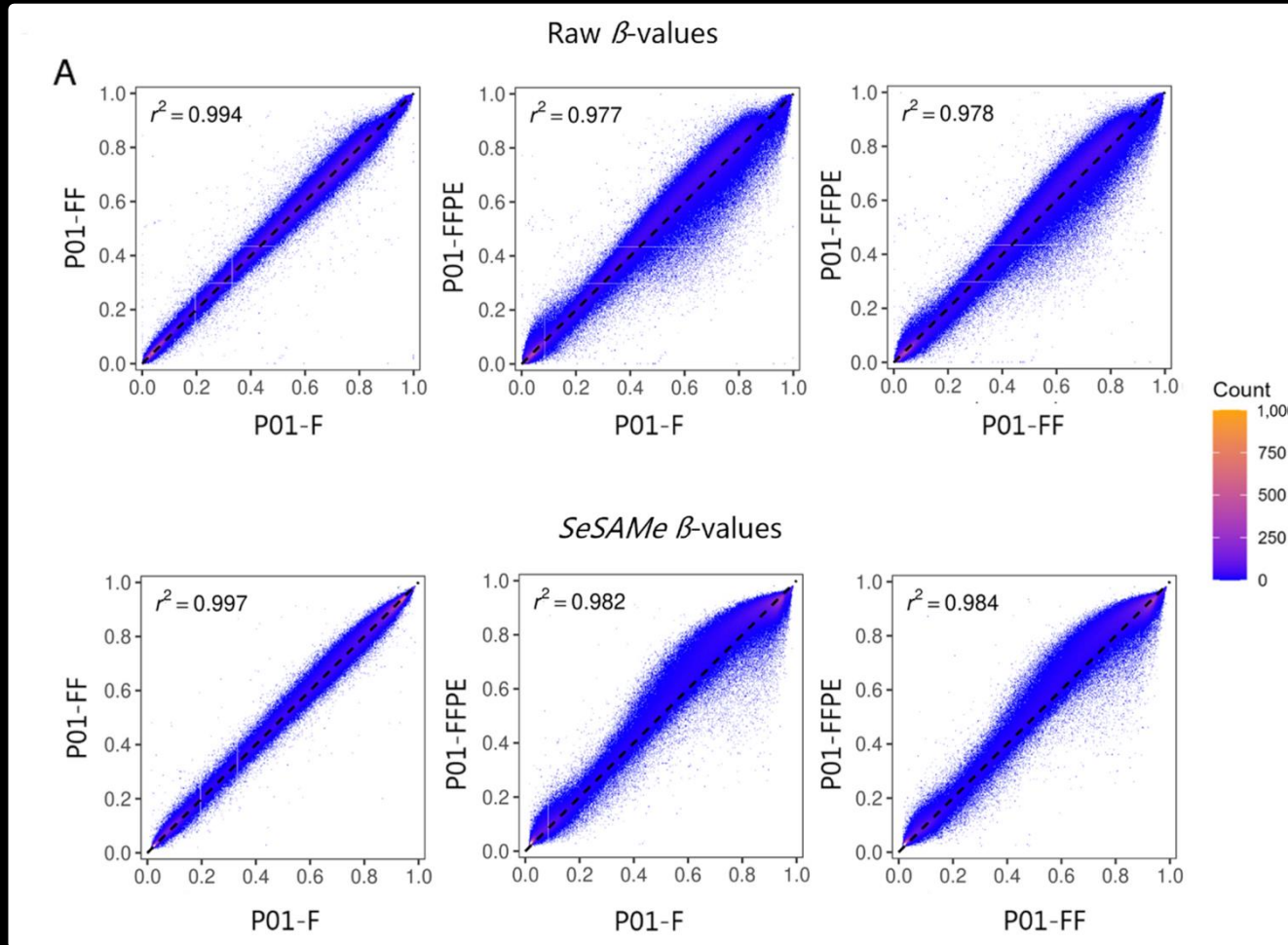
EXAMPLES

- What else can we do with it?
- Where is the field headed?

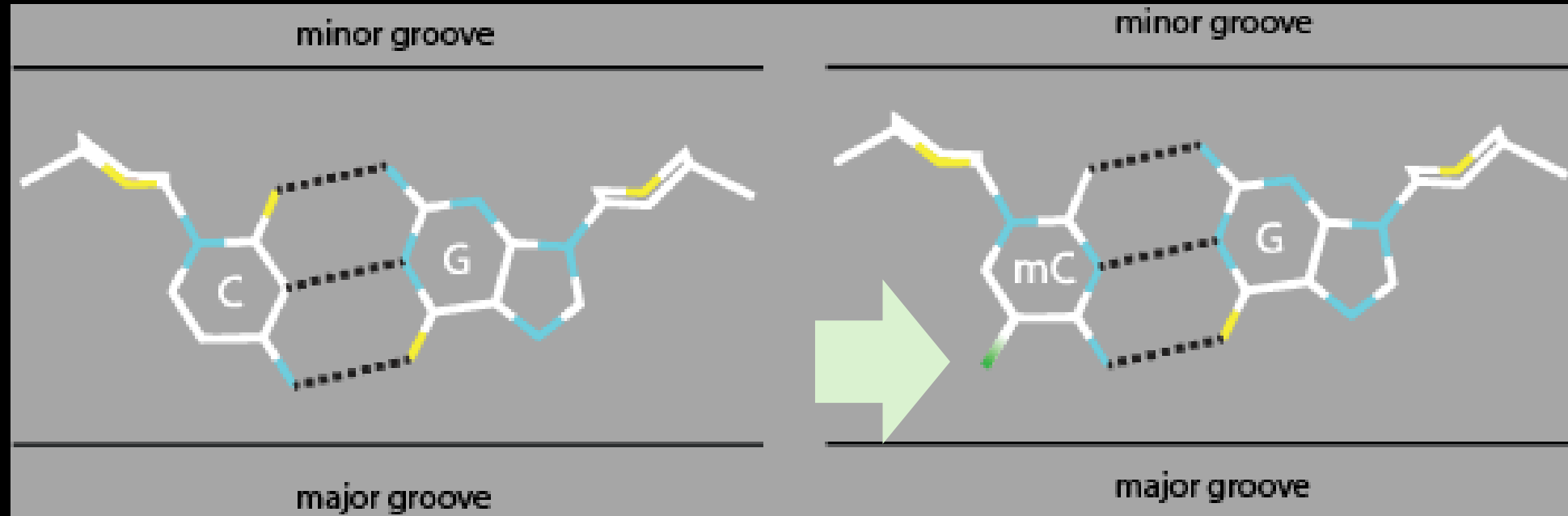
Methylation is stable in FFPE and specific across cancer staging



Methylation is stable in FFPE and specific across cancer staging



FOUNDATIONS OF DNA METHYLATION



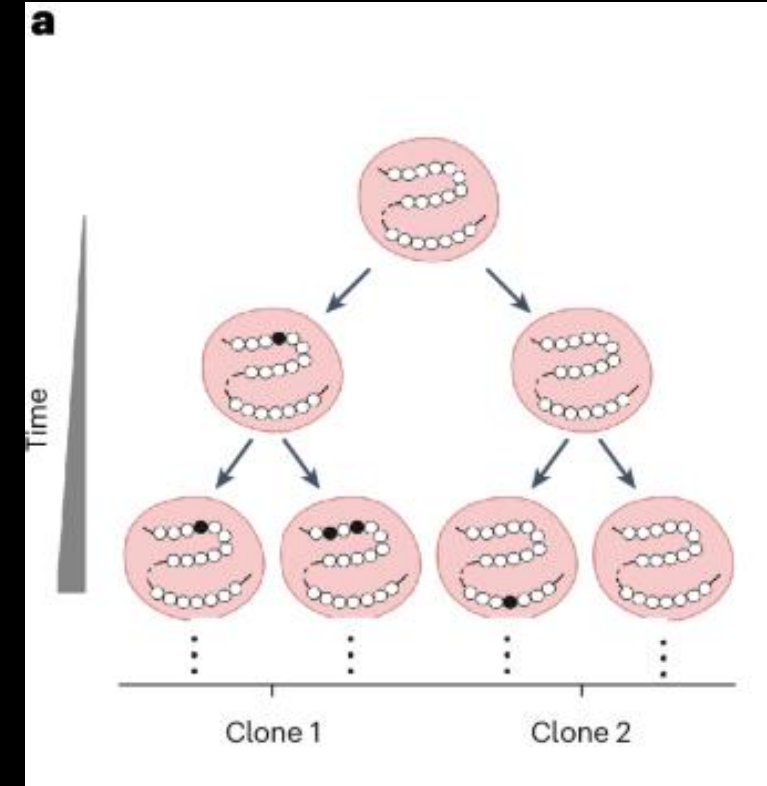
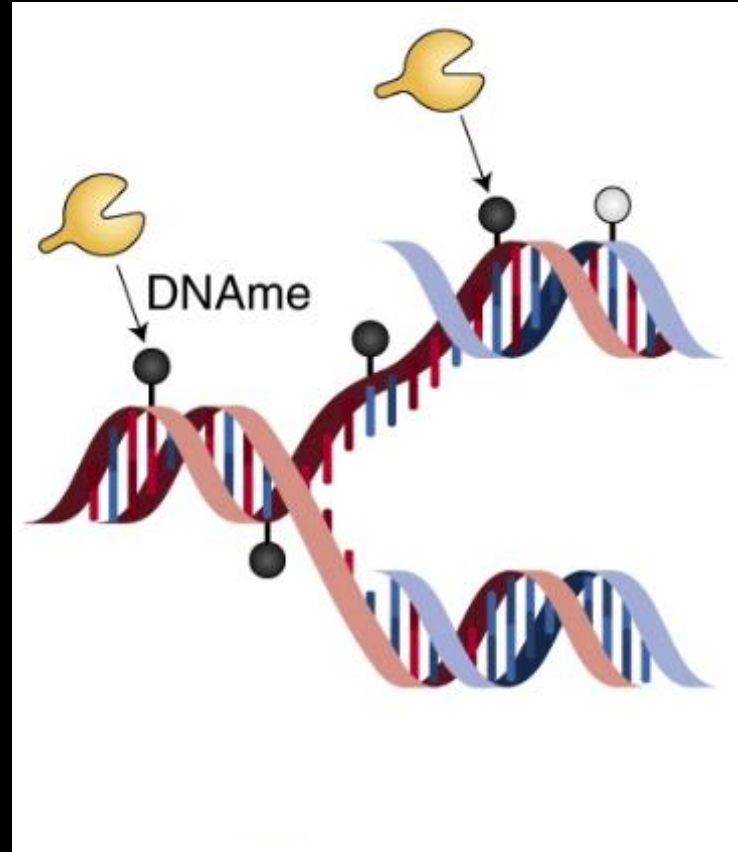
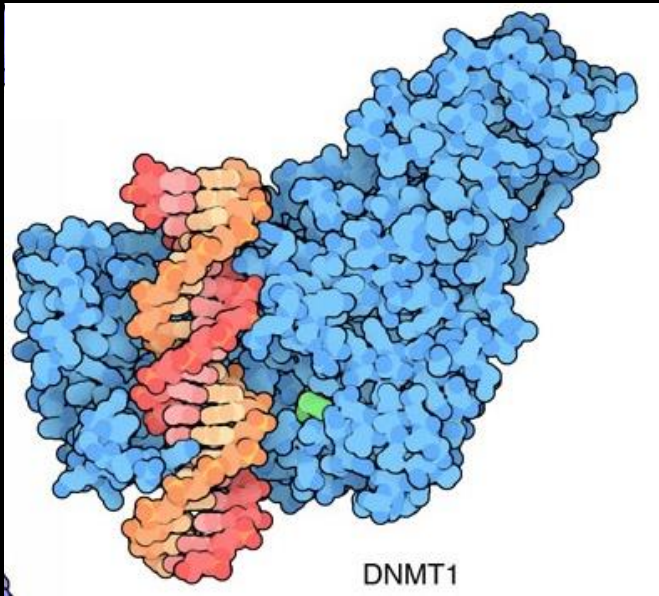
WRITER (2)

ERASER (1)

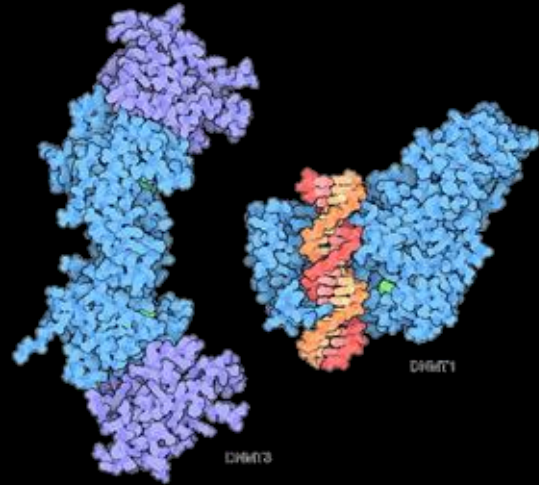
READER (4)

WRITER: DNMT1 (DNA MethylTransferase 1)

maintains methylation by strand symmetry
at CG dinucleotides



WRITER: DNMT3 (DNA MethylTransferase 3): de novo methylation on closed chromatin

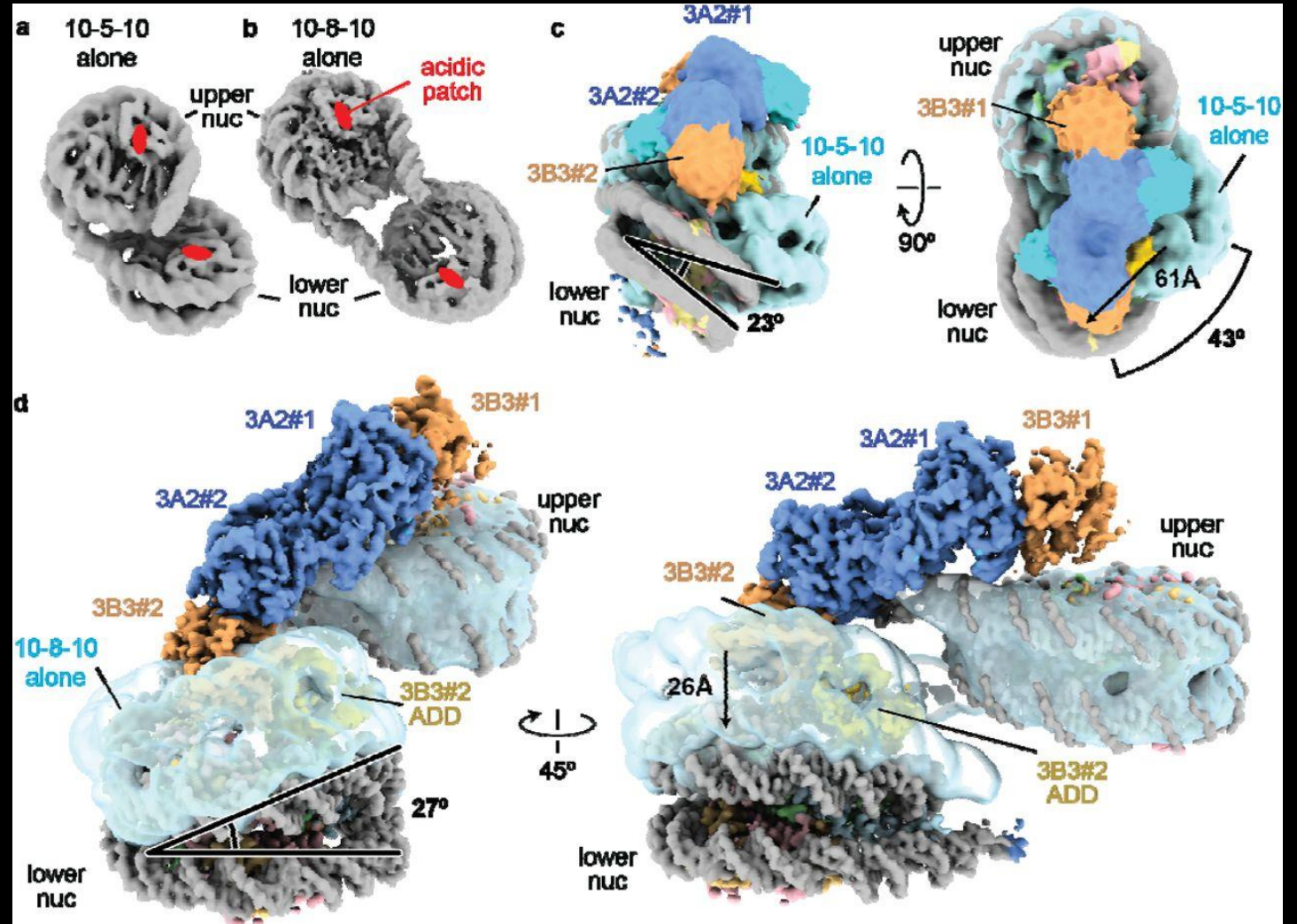
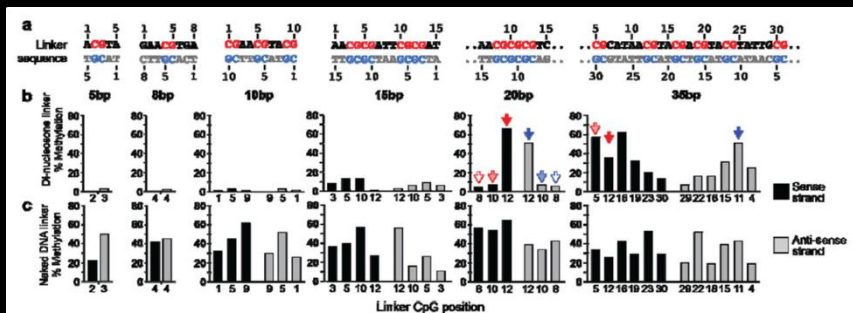


DNMT3

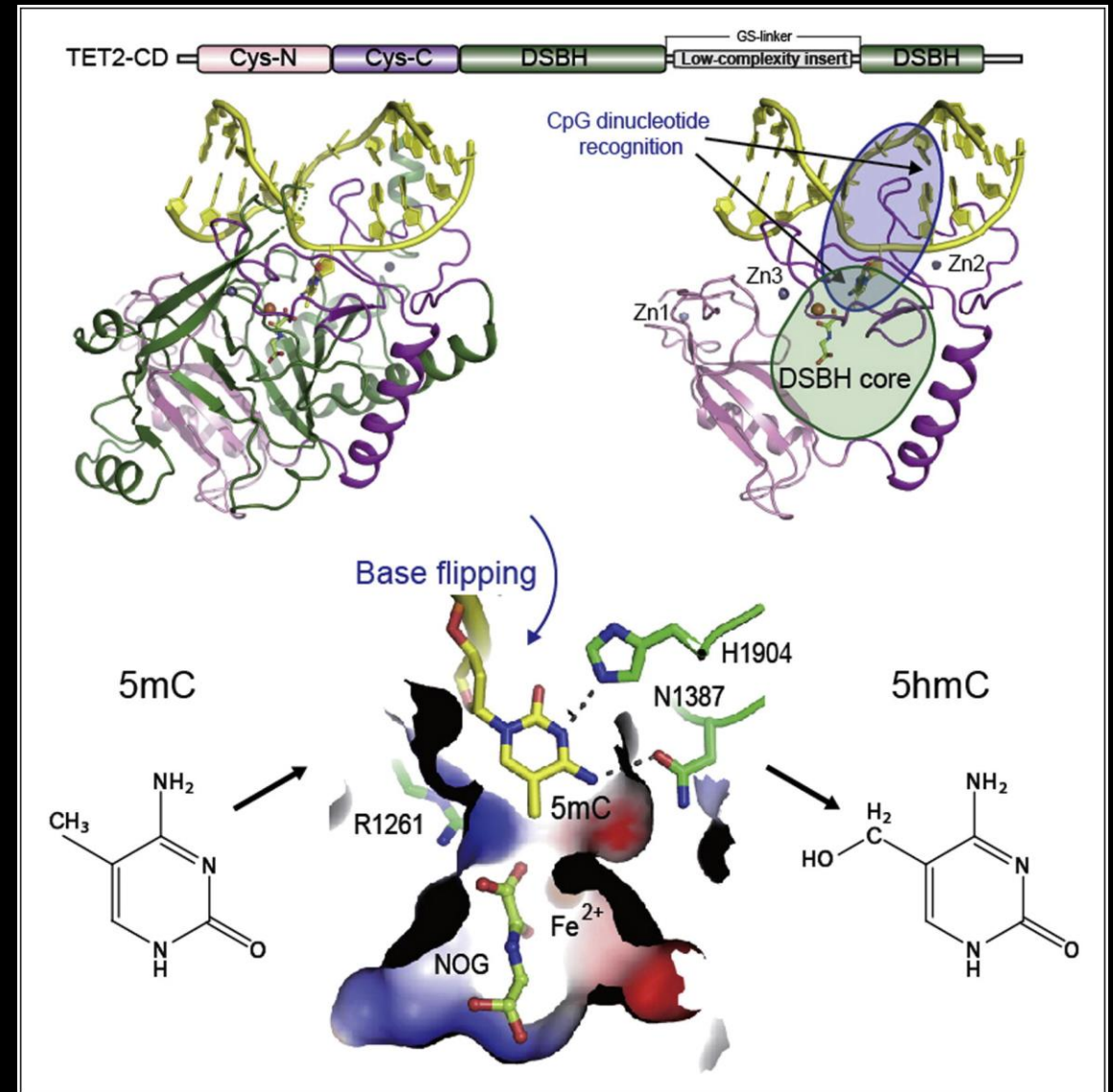
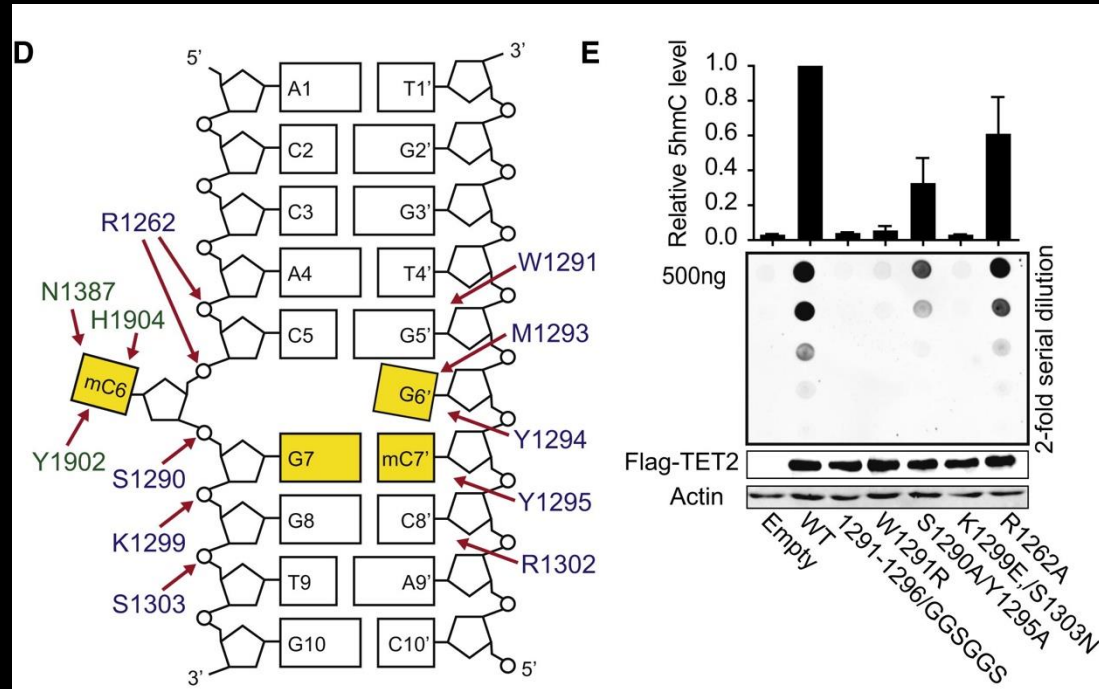
binds dinucleosomes marked with H3K36me3

methylates within the linker region > 15bp

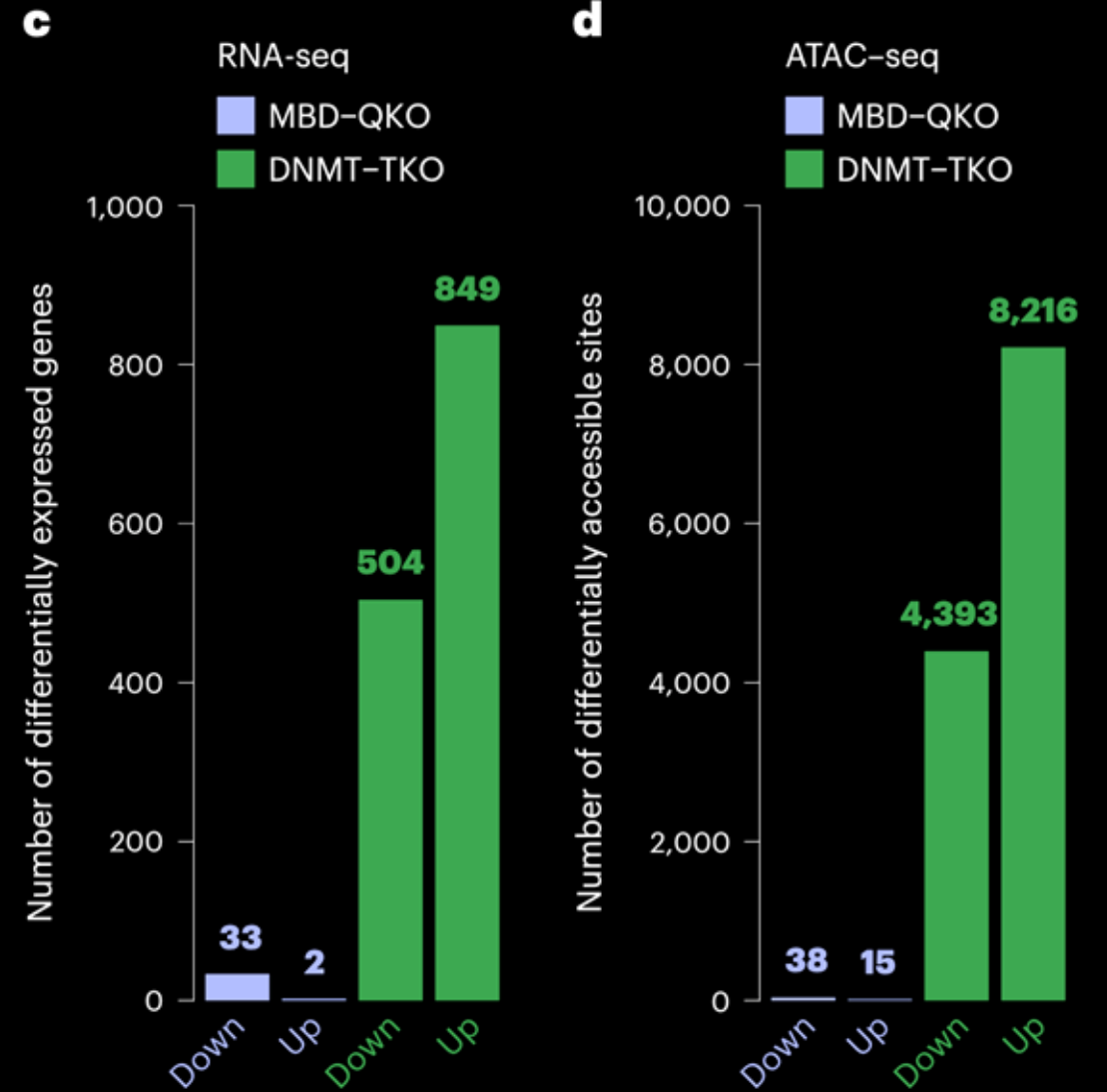
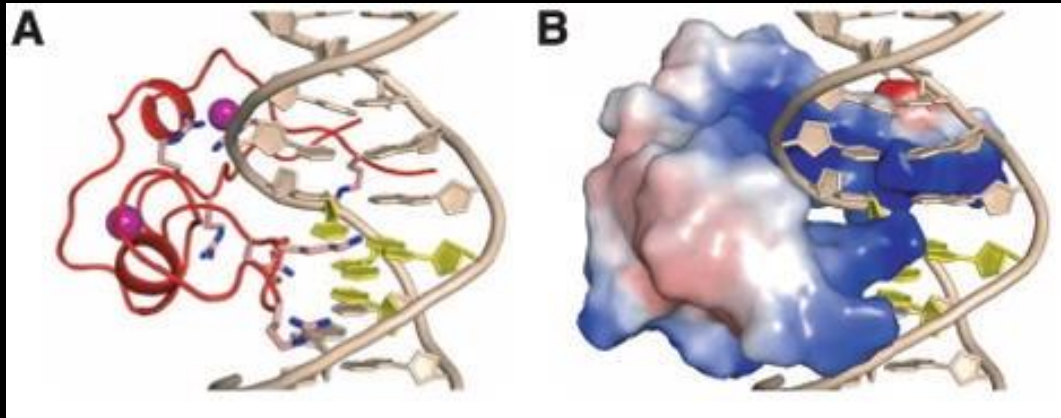
lack of MBD means DNMT3 can methylate non-CG dinucleotides
and can induce hemi-methylation



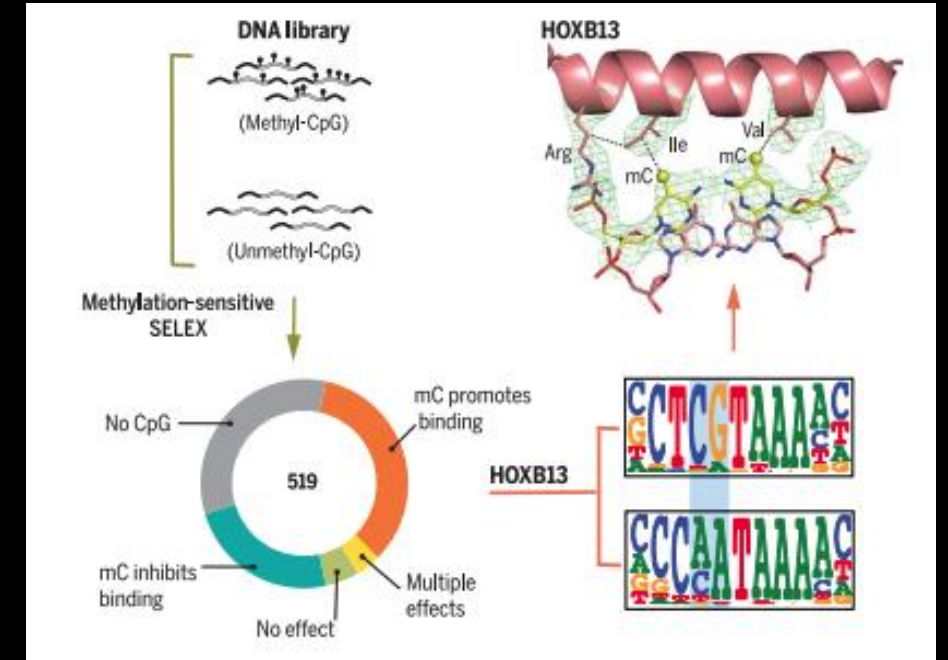
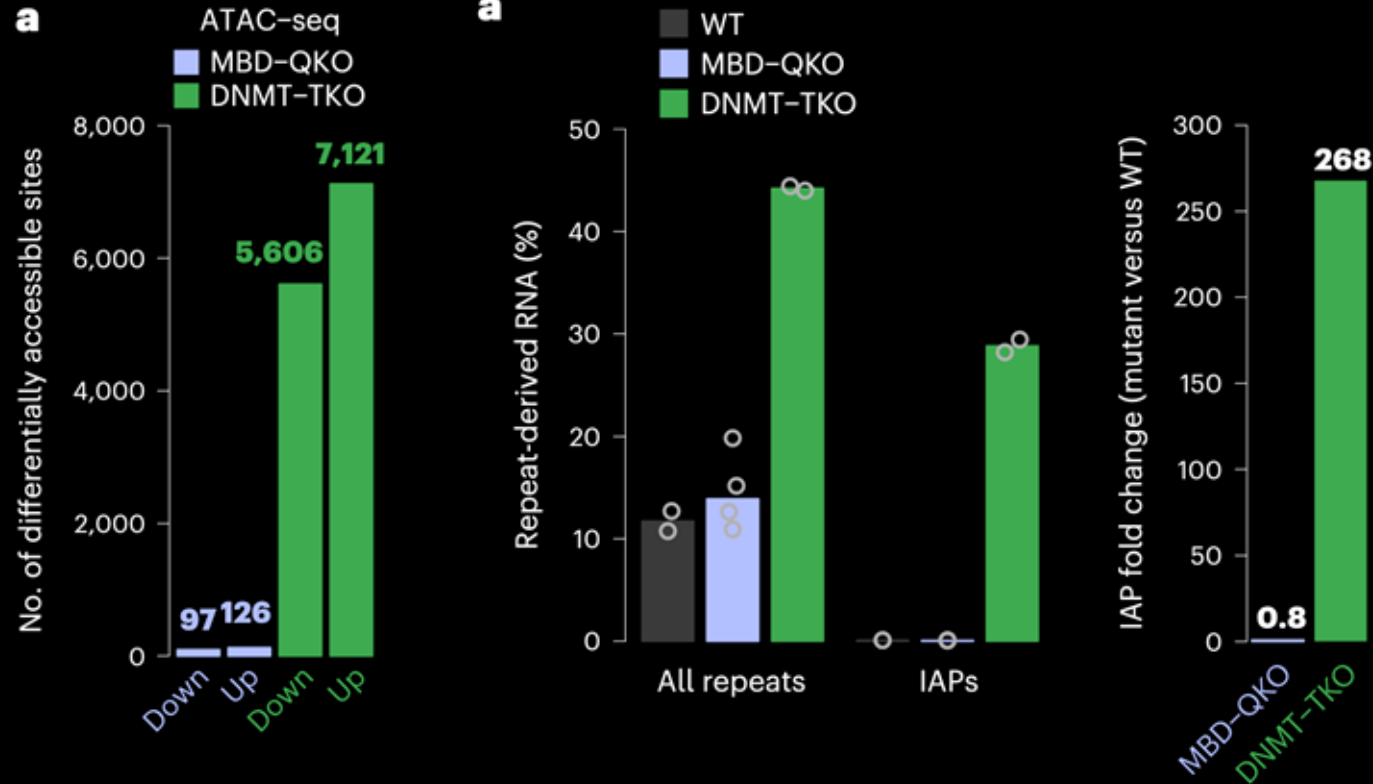
ERASER: TET2 (Ten Eleven Translocase 2) oxidizes 5mC marking it for base excision repair



READERS: Only 4 transcription factors have MethylBinding Domains (MBD)



READERS: >50% of transcription factors show CG-methylation preference

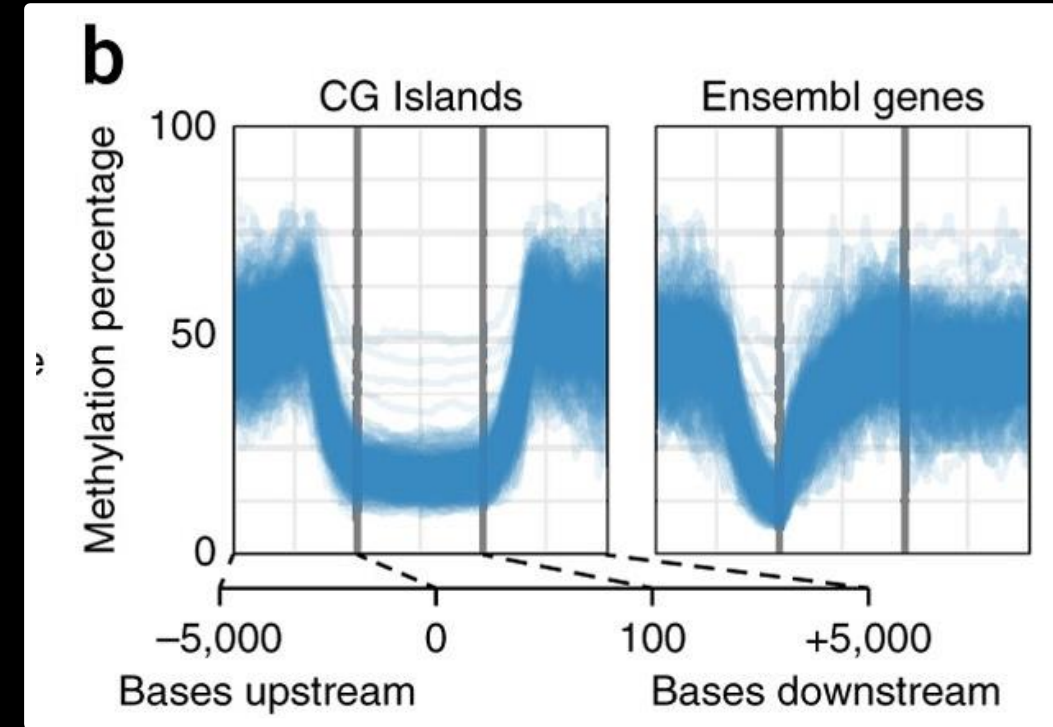
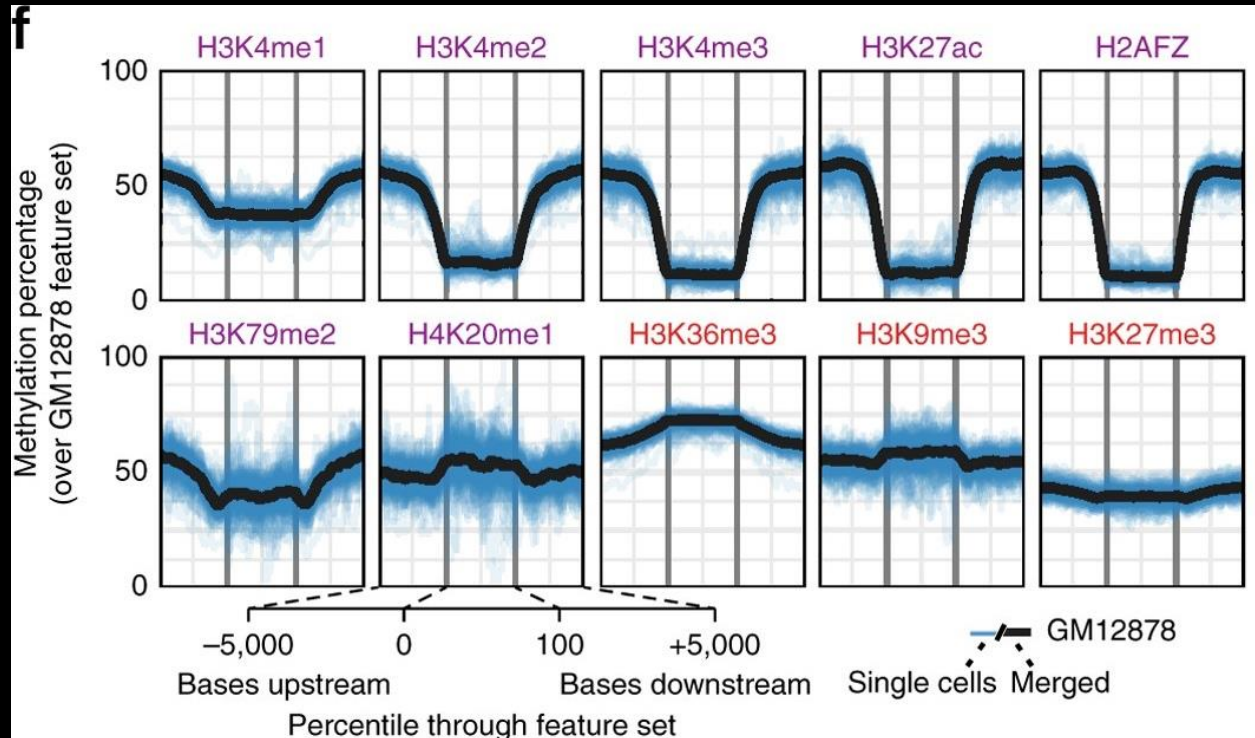


Yin, Yimeng, et al. *Science* 356.6337 (2017): eaaj2239.

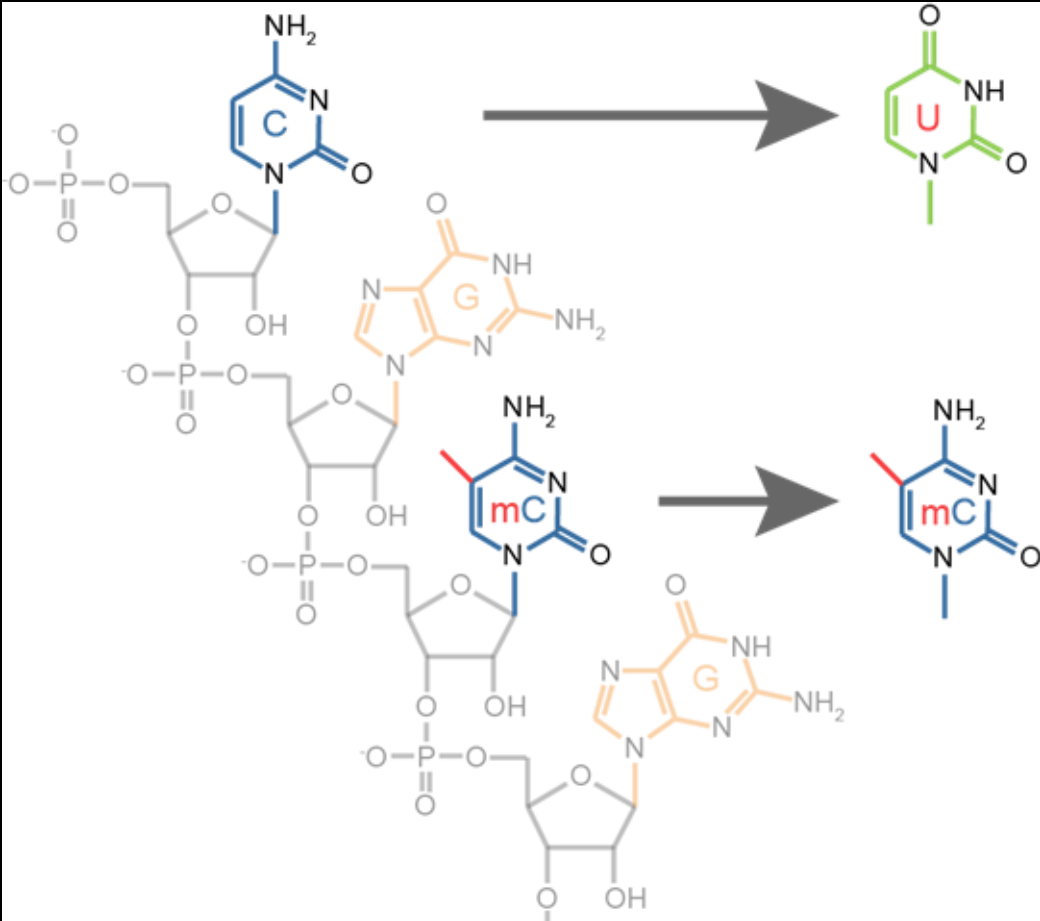
Kaluscha, Sebastian, et al. *Nature genetics* 54.12 (2022): 1895-1906.

METHYLATION ALONG THE GENOME:

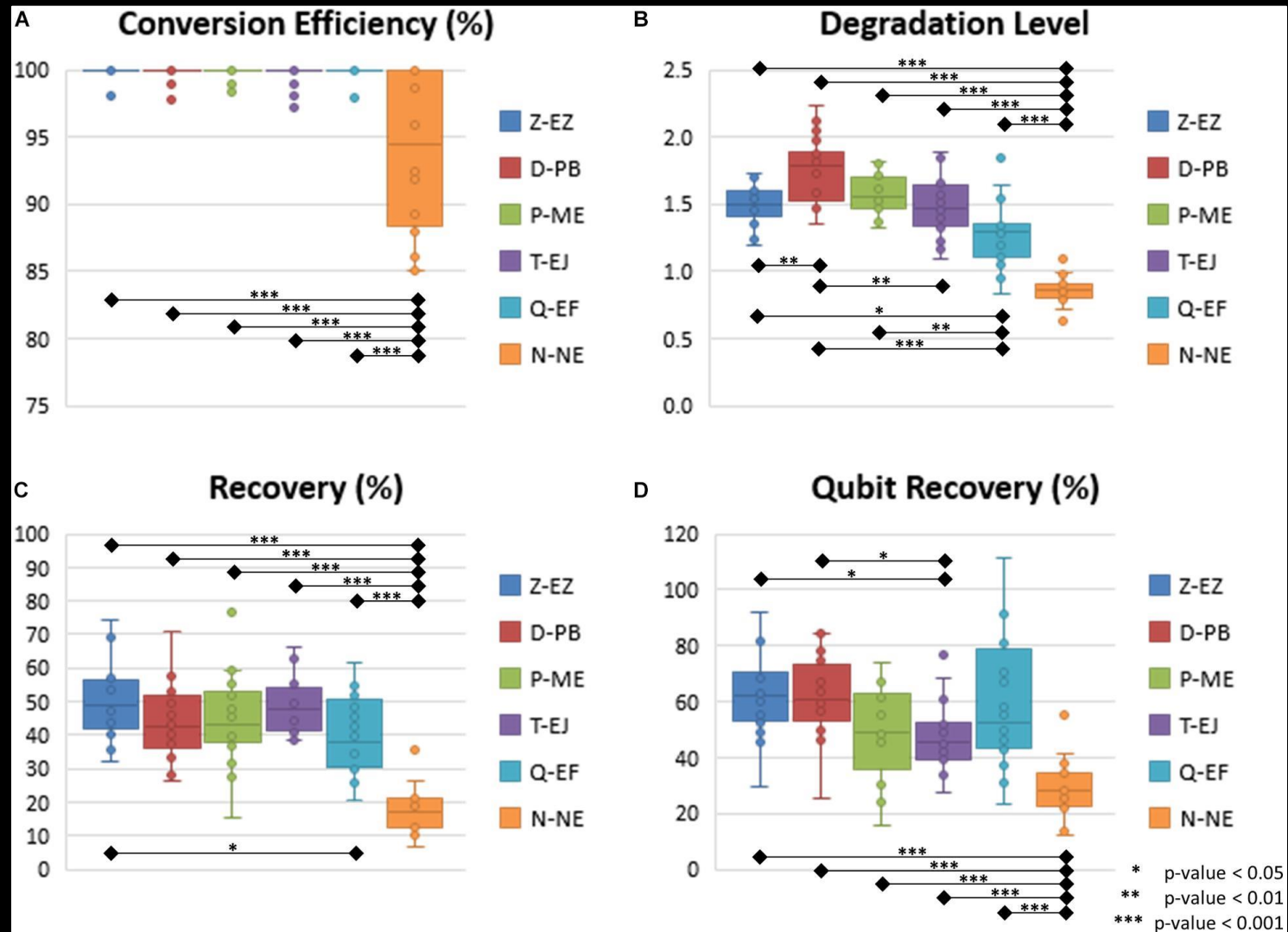
- CG sites are 60-80% methylated depending on cell type
- ▼ active sites ▲ silenced sites
- Clusters of CG sites are known as CG islands, these overlap promoters and enhancers and are ~80-90% hypomethylated



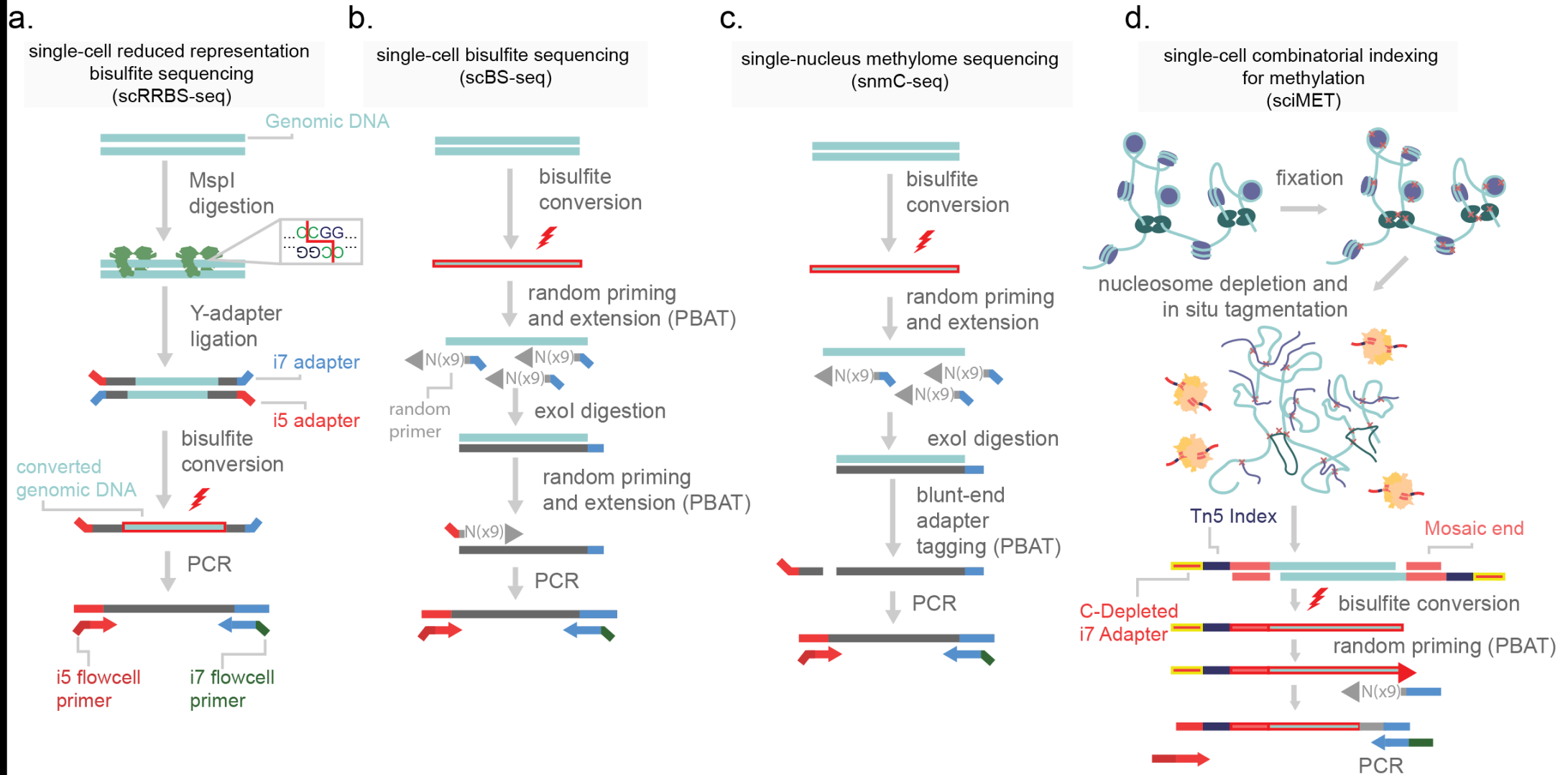
MEASURING METHYLATION THROUGH INDUCED POINT MUTATIONS



BISULFITE CONVERSION DEGRADES LIBRARIES



METHODS



STUDIES TO DATE



scMethBank

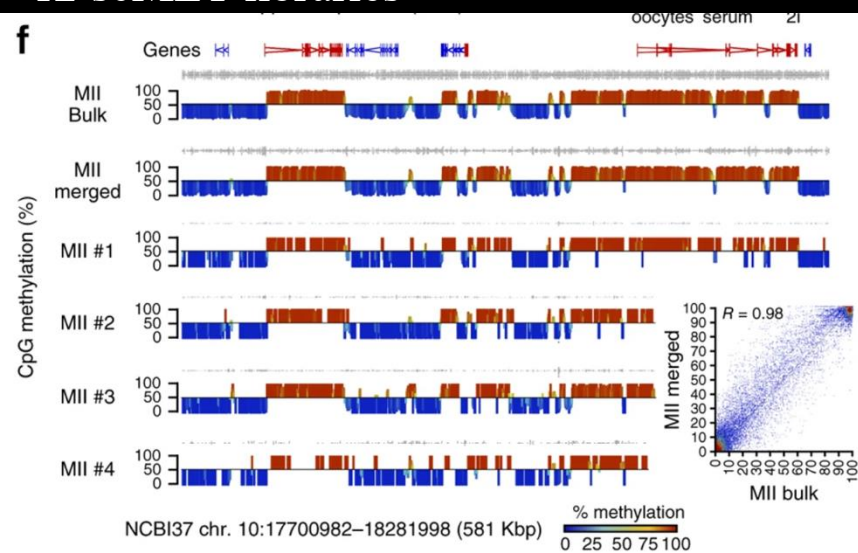
A database of single-cell methylation maps for human and mouse.

PMID	Sample	Year	Journal	Strategy	Cell Count
25042786	Mouse Oocytes	2014	Nat Methods.	scBS-seq	53
26549683	Mouse Oocytes	2015	Mol Cell	scBS-seq	27
27696608	Mouse Oocytes	2016	Science	scPBAT	10
26752769	Human Embryonic cells	2016	Genes Cells.	scM&T-seq	96
29255258	Human Embryonic cells	2017	-	scPBAT	506
28798132	Human and Mouse Cortex	2017	Science	snmC-seq	4,552
30078558	Human Hematopoietic Cells	2018	Stem Cell Reports.	scPBAL	219
30031774	Human Embryonic cells	2018	Cell Syst	scBS-seq	334
29915357	Human Embryonic cells	2018	Nat Cell Biol.	scCOOL-seq	285
30498128	Human Colorectal Cancer	2018	Nat Genet	scTrioSeq2	581
29644997	Cell Lines, Human and Mouse Cortex	2018	Nat. Biotechnology	sciMET	3,282
30560925	Mouse Oocytes	2019	Cell Res.	iscCOOL-seq	766
31554804	Mouse Muscle Stem Cells	2019	Nat Commun.	scBS-seq	260
-	Mouse Oocytes	2019	-	scM&T-seq	24
31847873	Mouse Oocytes	2019	Genome Med	scBS-seq	5
34616061	Mouse cortex	2021	Nature	snmC-seq	103,982
36494343	Human Cortex	2022	Nature Commun.	sciMETv2	2,546
39719707	Human Cortex	2025	Cell Genomics	sciMETv3	>140,000

HISTORY OF THROUGHPUT

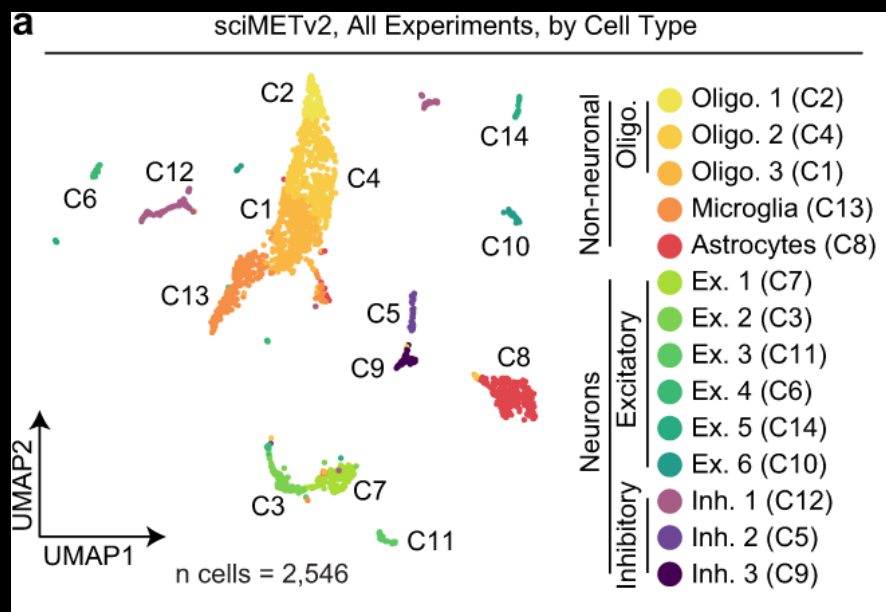
11 years ago

12 scMET libraries



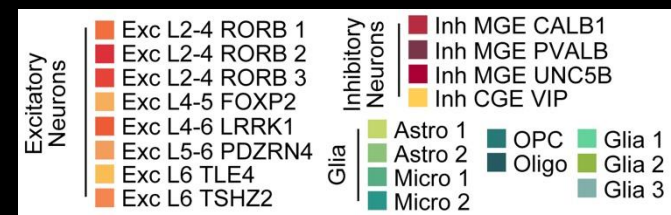
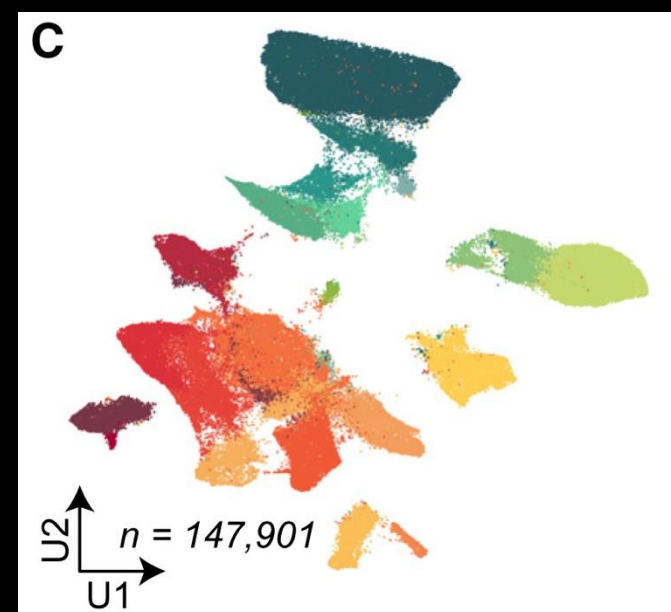
2022 sciMETv2

2,546 cells



2025 sciMETv3

147,901 cells

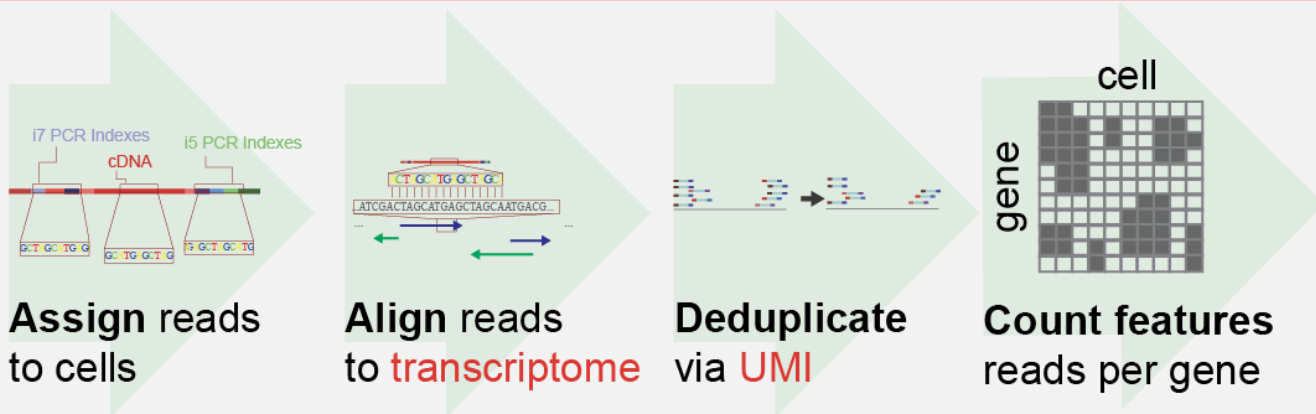


Smallwood et al. 2014

Nichols et al. 2022

Nichols et al. 2024

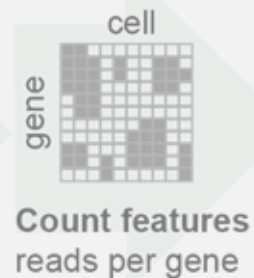
COUNTS BASED ASSAYS



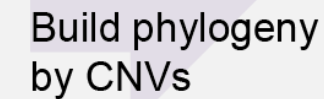
- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Gene network analysis

COUN

scWGS



- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Gene network analysis



- Features are genome bins instead of genes

scRNA
TRANSCRIPTION

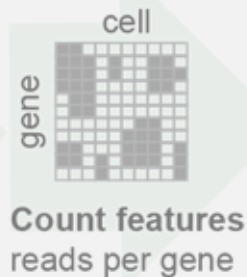
Assign reads
to cells



Align reads
to **transcriptome**



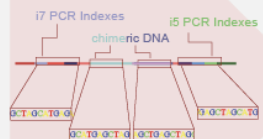
Deduplicate
via **UMI**



Count features
reads per gene

- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Gene network analysis

E

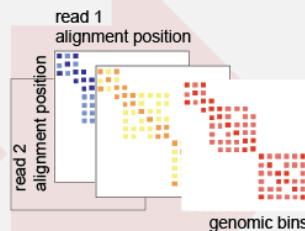


Assign reads
to cells

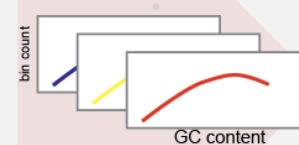


Align **paired end**
reads to genome

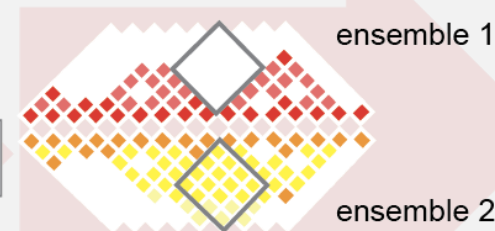
Deduplicate
PCR reads



Count features
read ends per
2D bin



Correct bins
by GC content



Define
topological domains

- Read 1 and Read 2 of Paired end are aligned separately giving a 2D genome bin

scRNA
TRANSCRIPTION

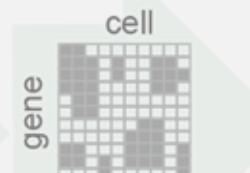
Assign reads
to cells



Align reads
to **transcriptome**



Deduplicate
via **UMI**



Count features
reads per gene

- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Gene network analysis

scATAC



Assign reads
to cells



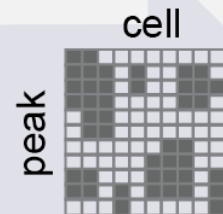
Align reads
to genome



Deduplicate
PCR reads



Find features
by **read pileup**



Count features
reads per peak

- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Transcription factor motifs
- Promoter-Enhancer linkage

scHiC
CHROMATIN
CONFORMATION

Assign reads
to cells



Align **paired end**
reads to genome



Deduplicate
PCR reads



Count features
read ends per
2D bin



Correct bins
by GC content



Define
topological domains

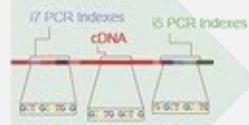


ensemble 2

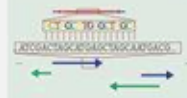
- Features of read pileups
(Open regions)

COUNTS BASED ASSAYS

scRNA TRANSCRIPTION



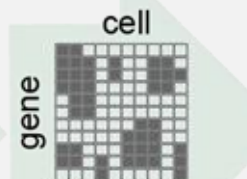
Assign reads to cells



Align reads to **transcriptome**



Deduplicate PCR reads via **UMI**



Count features reads per gene

- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Gene network analysis

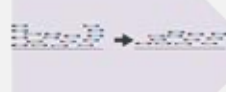
scWGS WHOLE GENOME



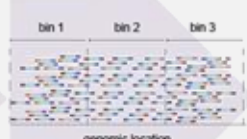
Assign reads to cells



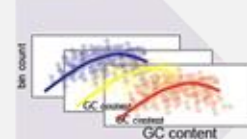
Align reads to genome



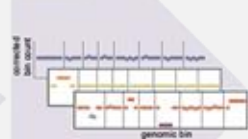
Deduplicate PCR reads



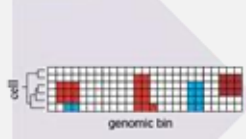
Count features reads per **1D bin**



Correct bins by GC content

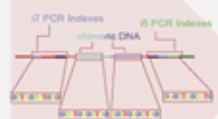


Call CNVs bins with $\pm$ expected reads



Build phylogeny by CNVs

scHiC CHROMATIN CONFORMATION



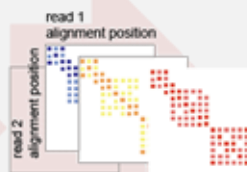
Assign reads to cells



Align **paired end** reads to genome



Deduplicate PCR reads



Count features read ends per **2D bin**



Correct bins by GC content

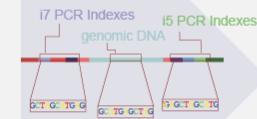


Define topological domains



ensemble 1
ensemble 2

scATAC CHROMATIN ACCESSIBILITY



Assign reads to cells



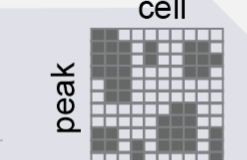
Align reads to genome



Deduplicate PCR reads



Find features by **read pileup**

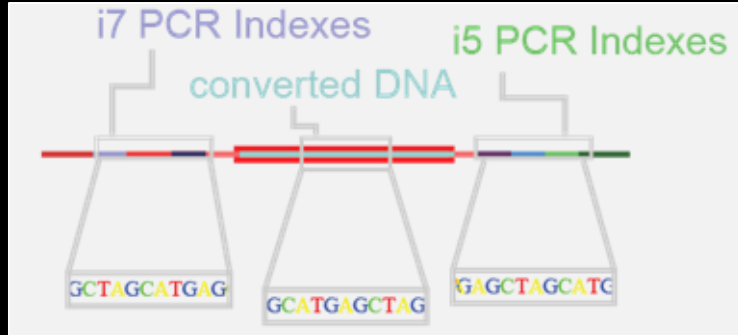


Count features reads per peak

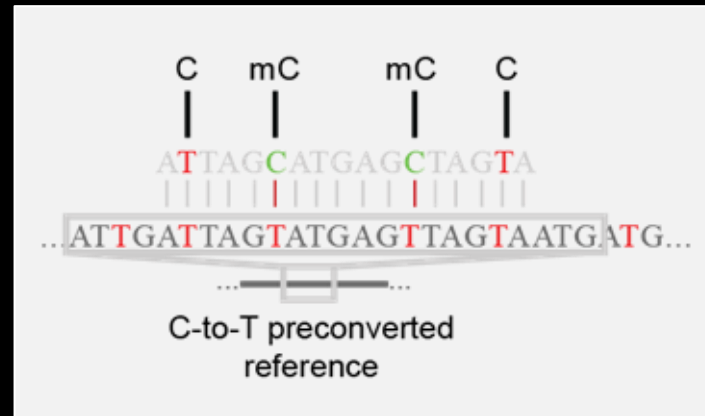
- Group cells by similarity
- Trajectory analysis
- Differential analysis
- Transcription factor motifs
- Promoter-Enhancer linkage

METHYLATION ALIGNMENT ► FEATURE SUMMARY ► CELL X FEATURE MATRIX

assign bisulfite converted reads to cells



identify methylated cytosines per cell

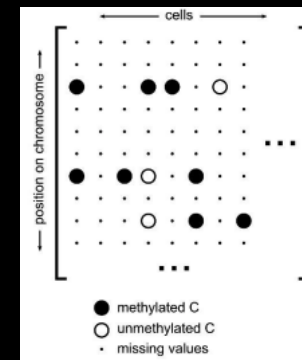


Convert cell data into a cell by feature matrix

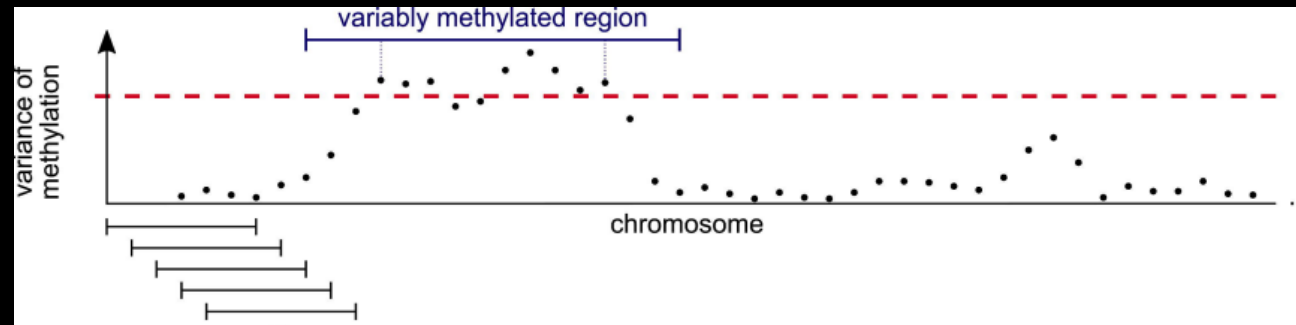
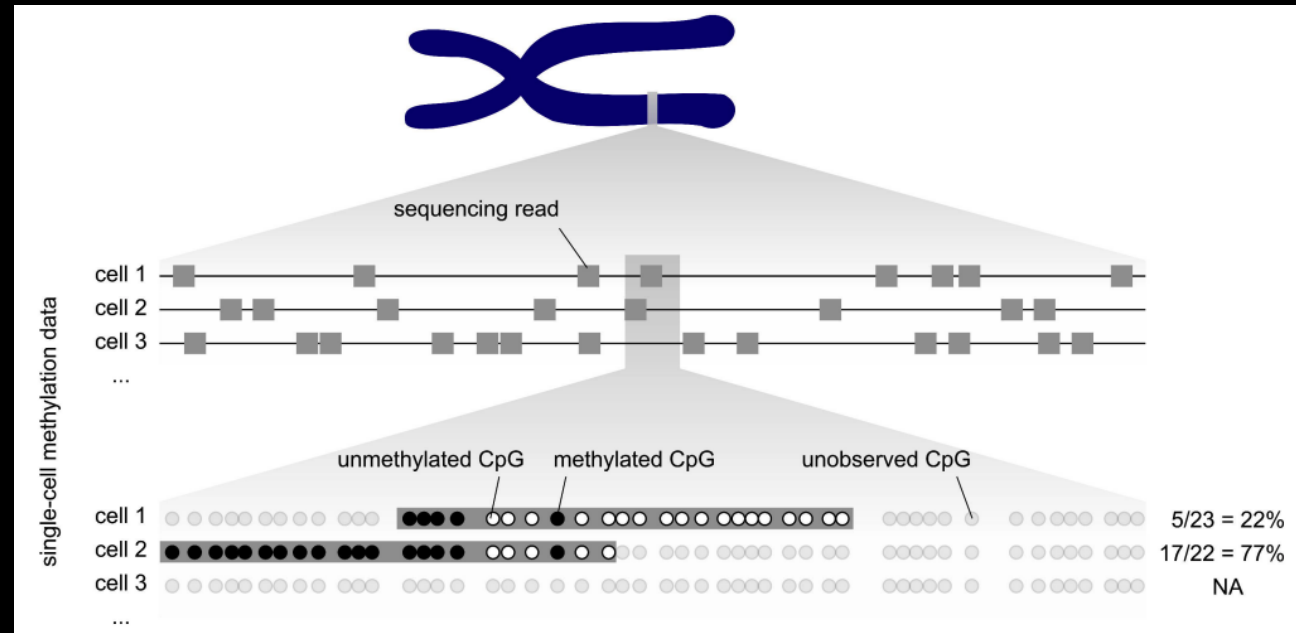
Aligners like bismark preconvert the reference genome and your data to increase alignment rate

methylation files

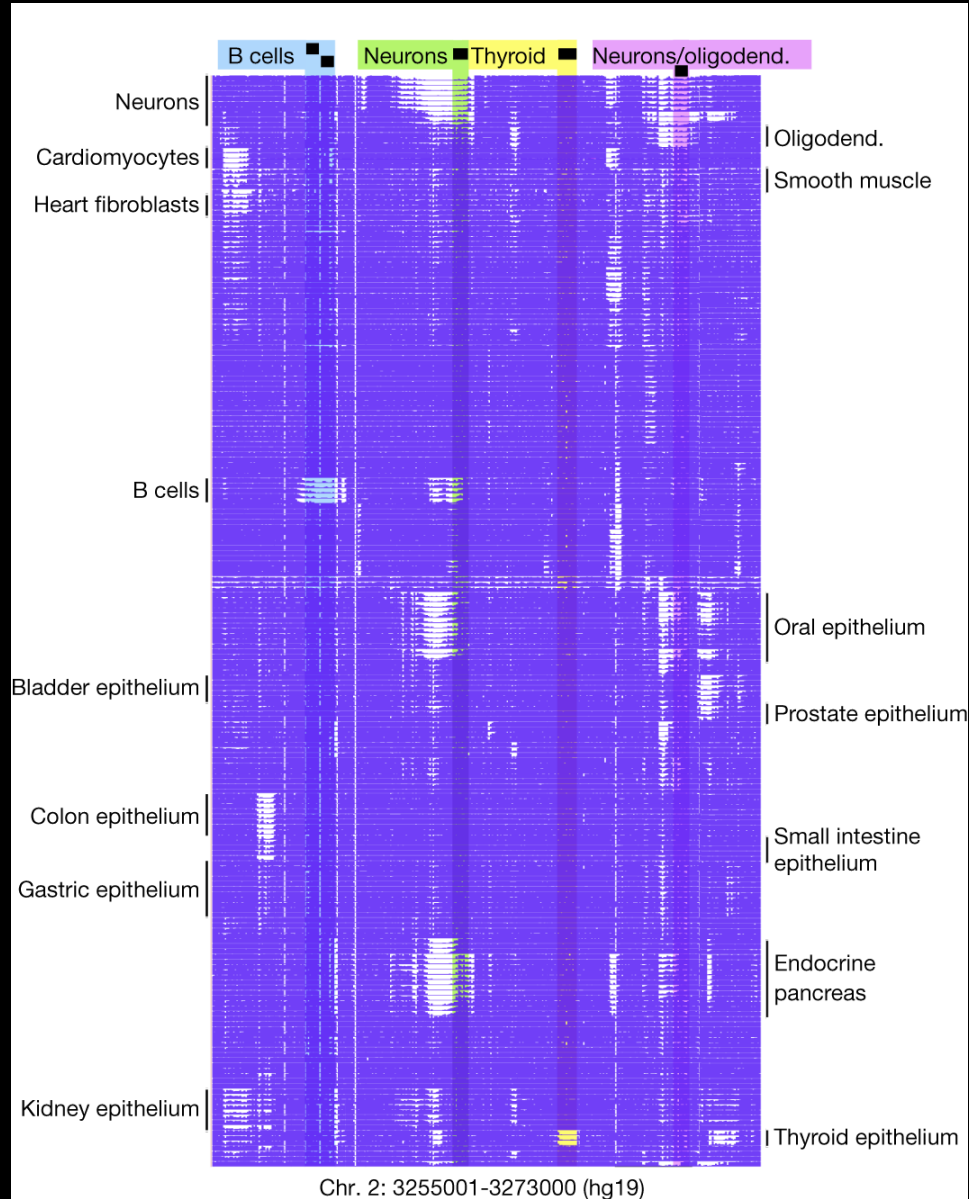
chrom	position	meth
chr1	1015303	•
chr1	1015314	•
chr1	1015319	•
chr1	1016521	o
chr1	1016533	o
chr1	1017587	•
chr1	1017589	o
chr1	1017611	•
chr1	1022010	•
chr1	1022019	o
...	...	...



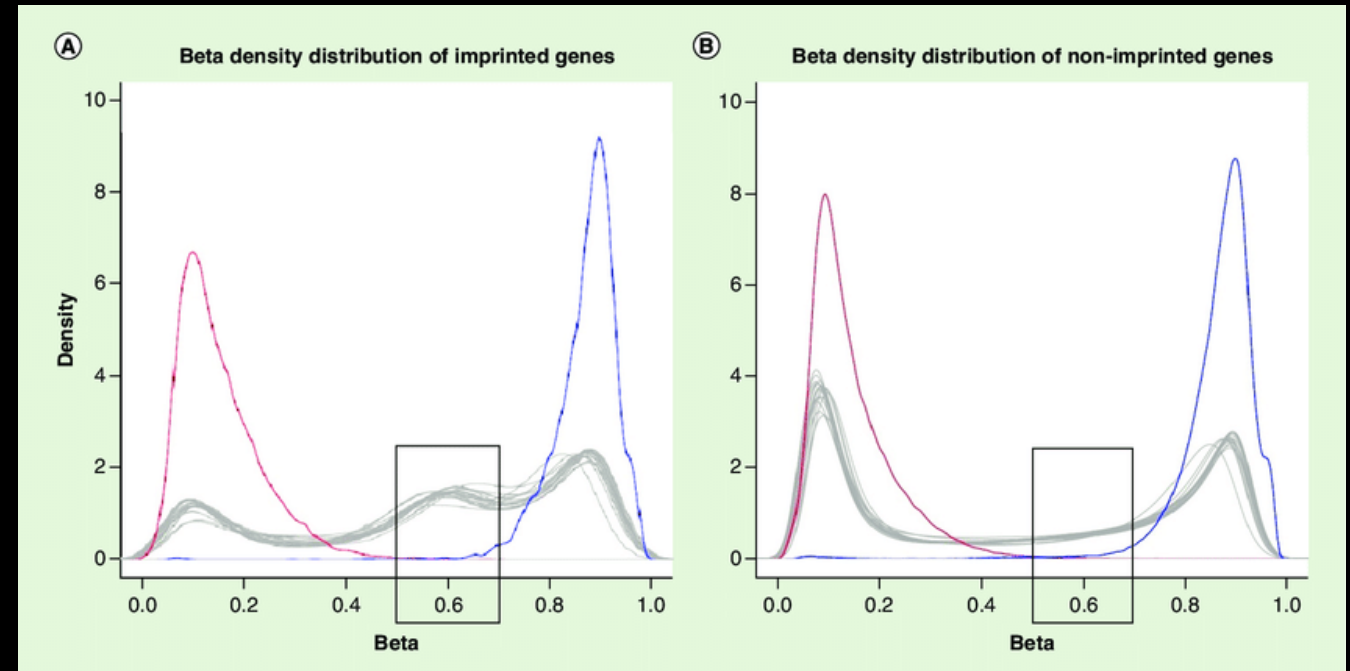
Unobserved regions are fundamentally different than counts based assays



Unobserved regions are fundamentally different than counts based assays



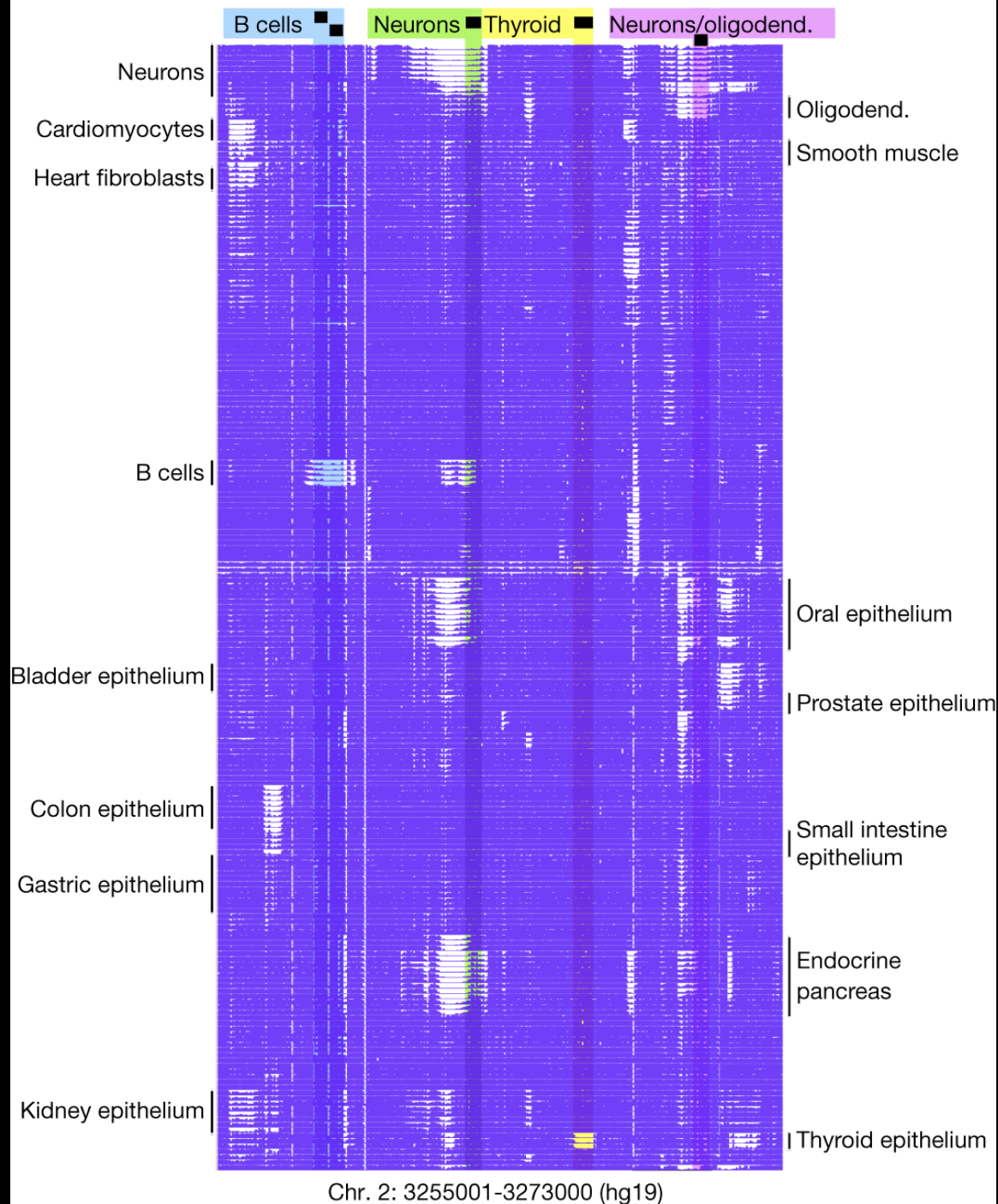
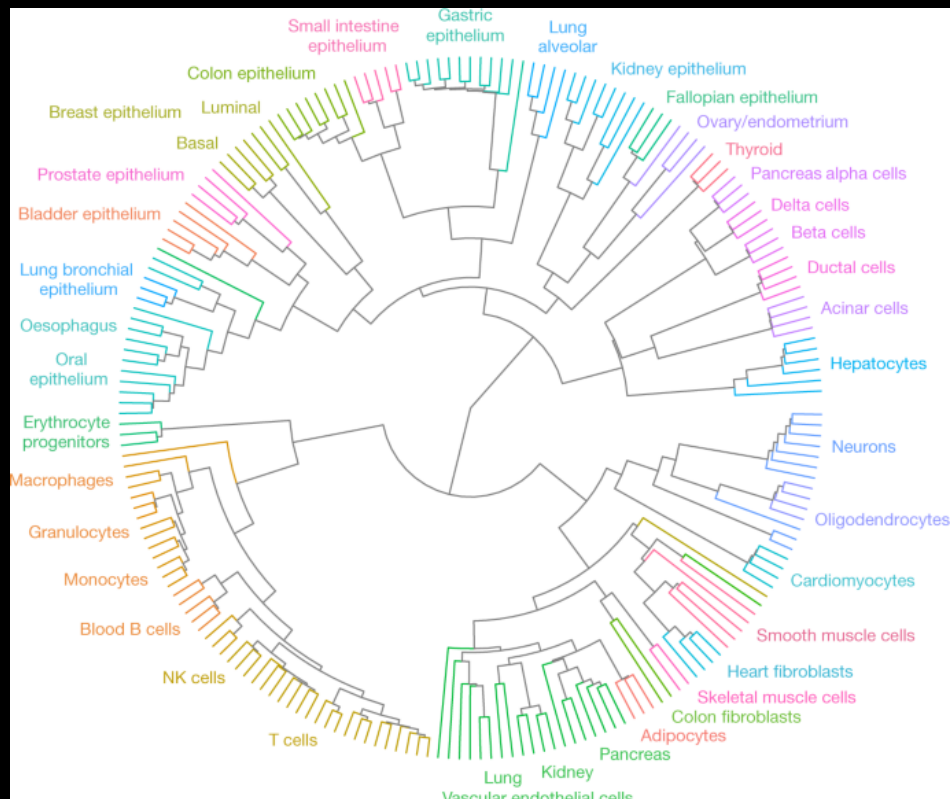
Methylation is binary in small genomic windows



15kbp locus

Cell typing!

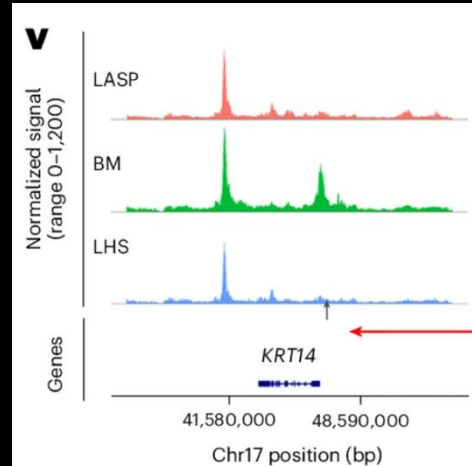
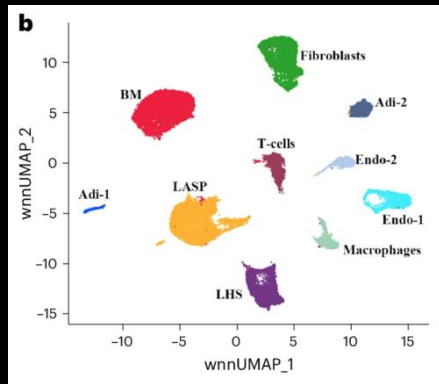
Bulk methylomes from purified cell types



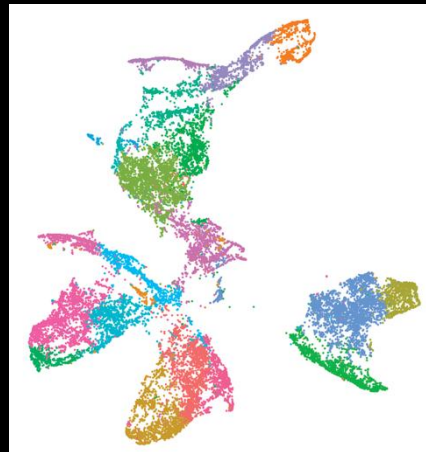
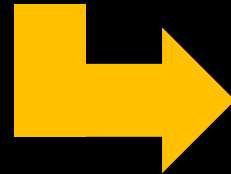
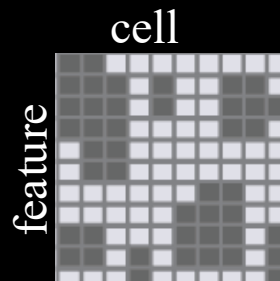
Chr. 2: 3255001-3273000 (hg19)

METHYLATION CLUSTER CELLS ON WINDOWS ► IDENTIFY DMRS ► RECLUSTER CELLS ON DMRS

Select starting features:
5kb bins
ATAC peaks
RNA promoters
CUT&TAG markers

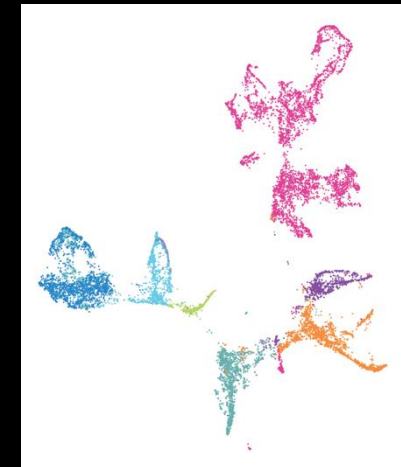


Weighted methylation
across features

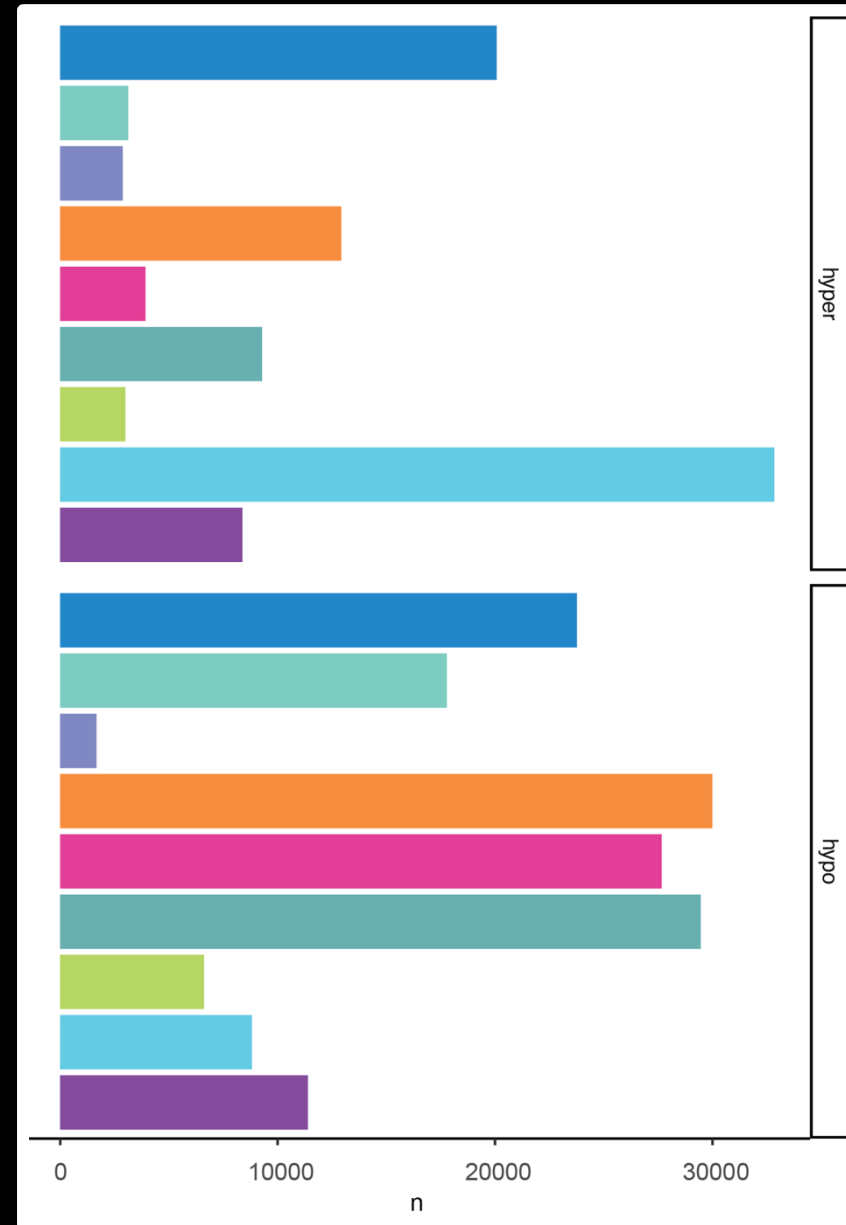


- cluster cells on these features
- use cell groupings to define DMRS

Cluster on DMRS and assign cell types



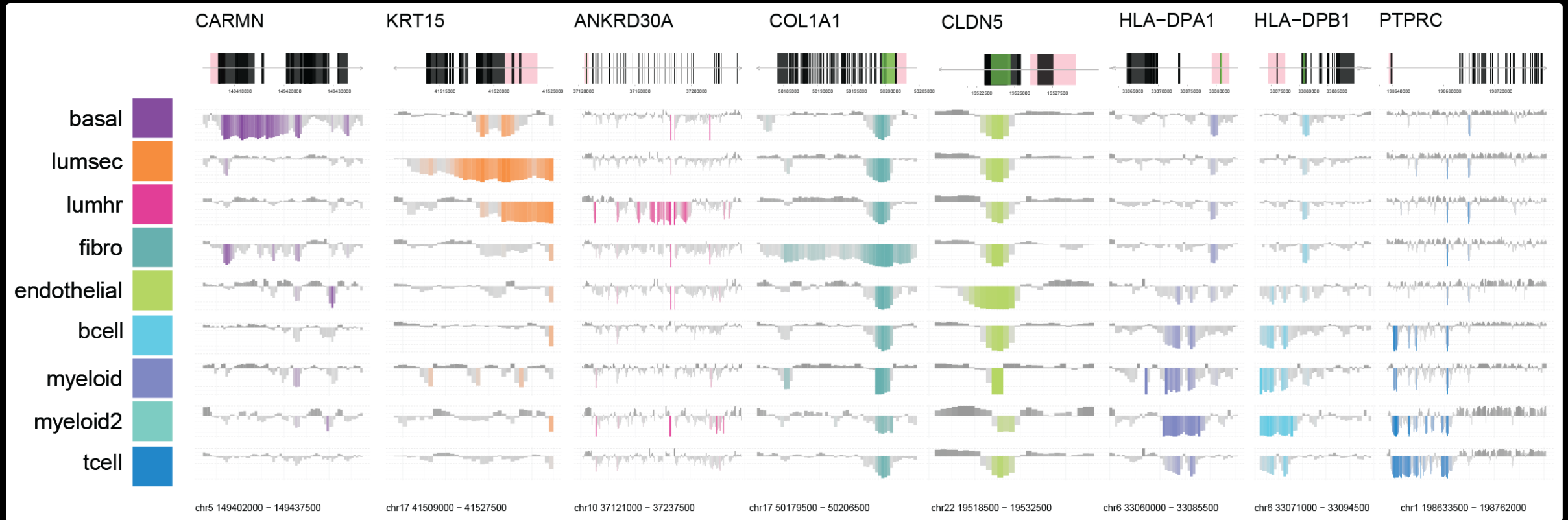
From cluster to cell types: *a priori*



96,436 DMRs
mean 3.6kbp

157,143 DMRs
mean 3.9kbp

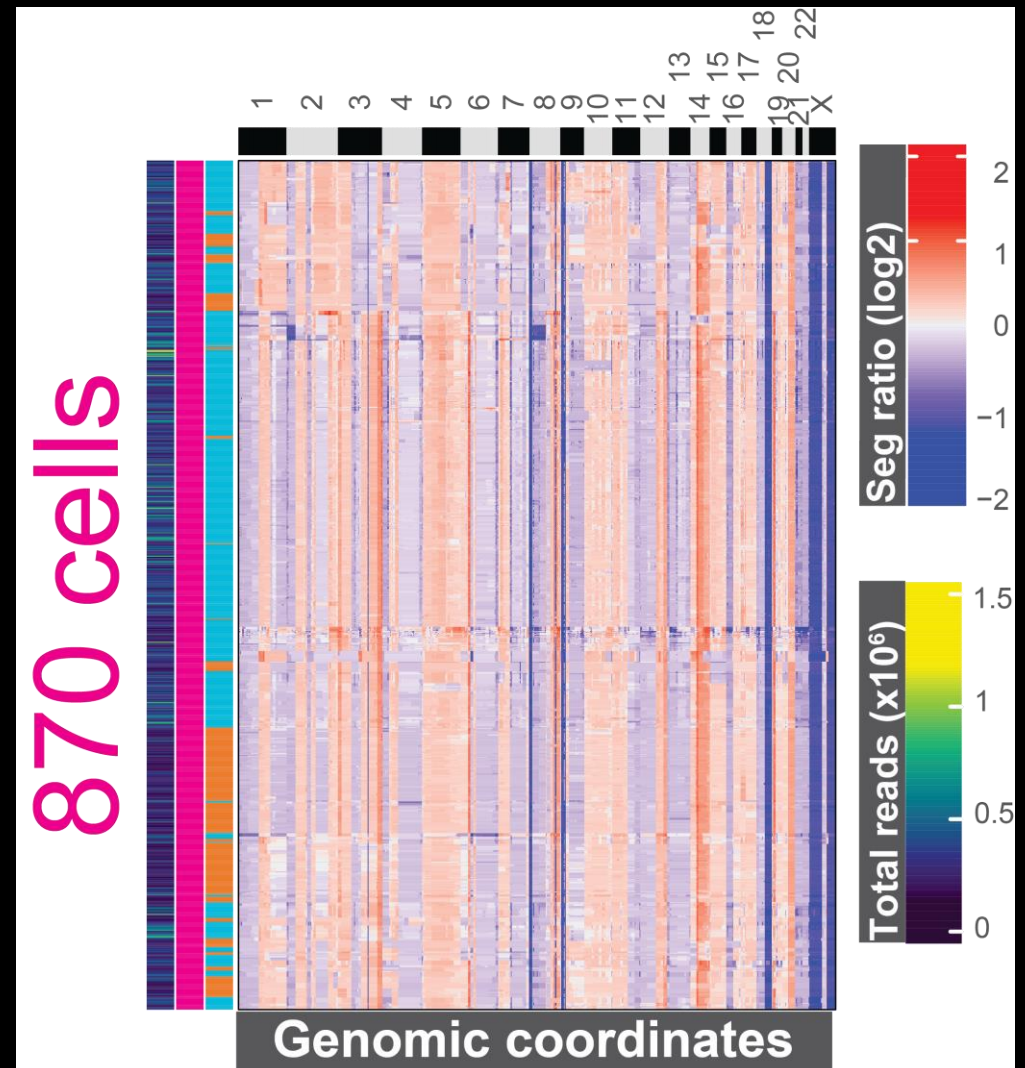
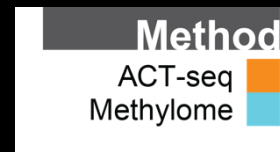
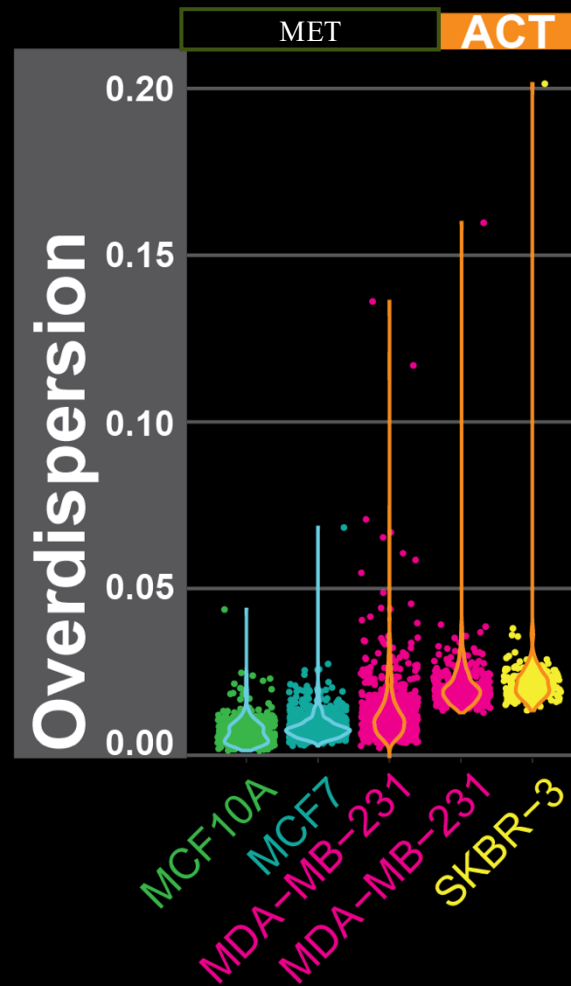
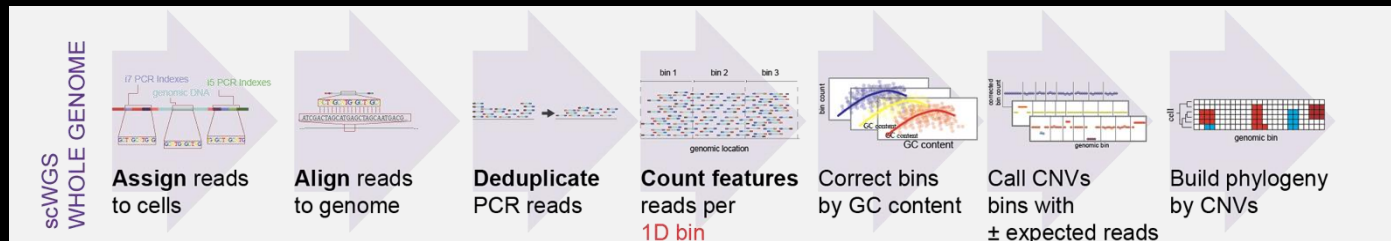
Cell typing in methylation: *a posteriori*



hyper  mean %mCG

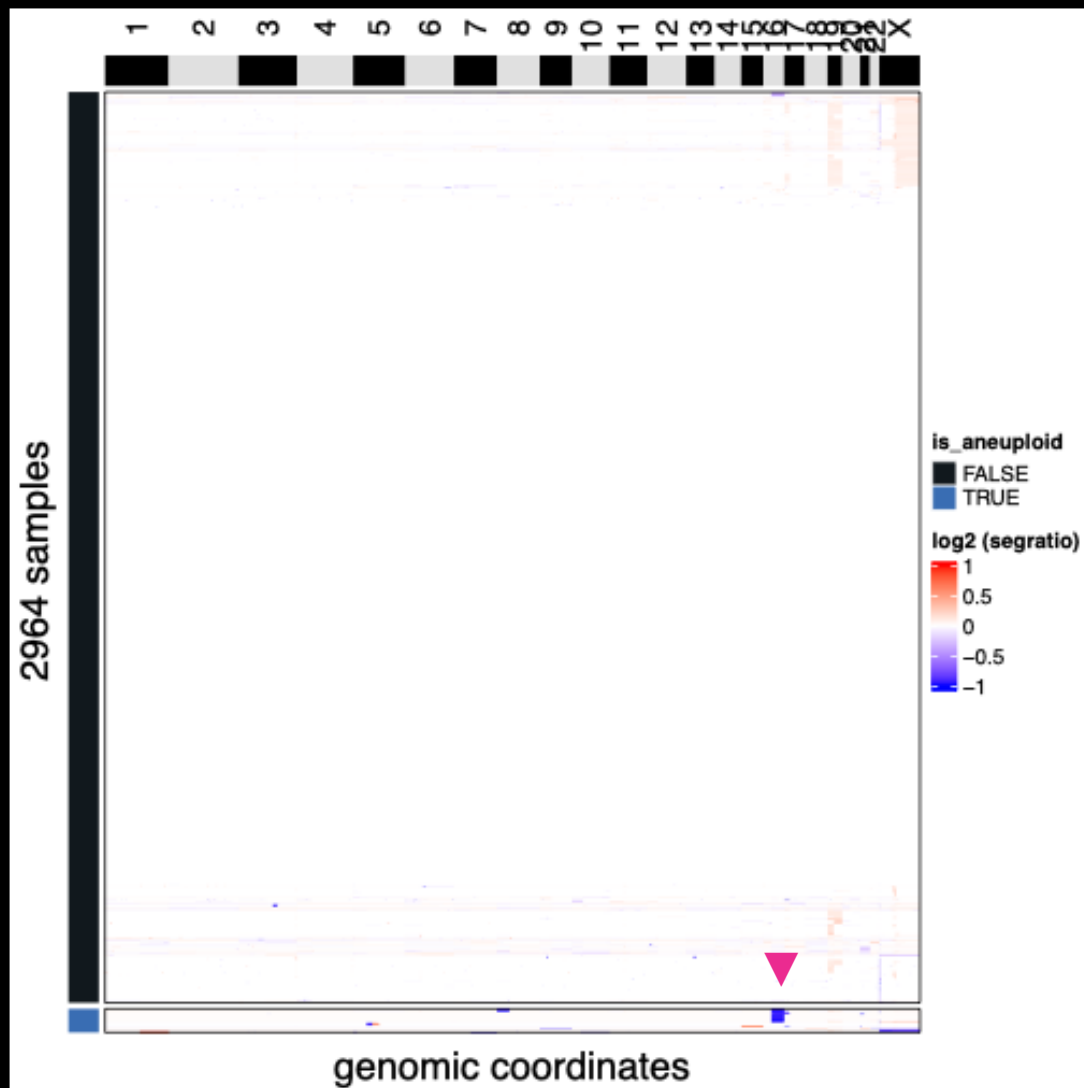
 hypo 500bp windows

Validating CNV calling on heterogenous cell lines



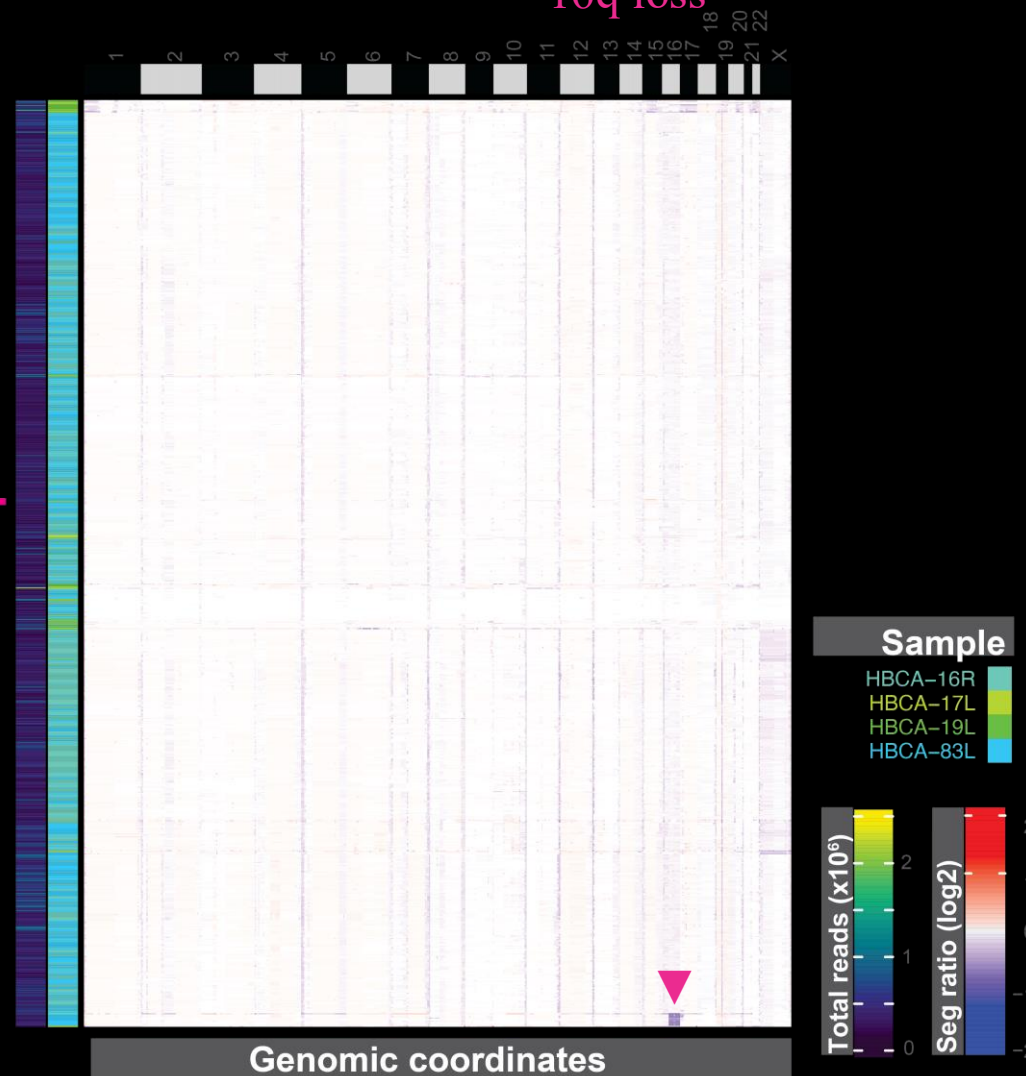
Lineage tracing through CNVs

icell8: Epithelial enriched



In normal breast tissue we found 1
CNV across 4 samples
16q loss

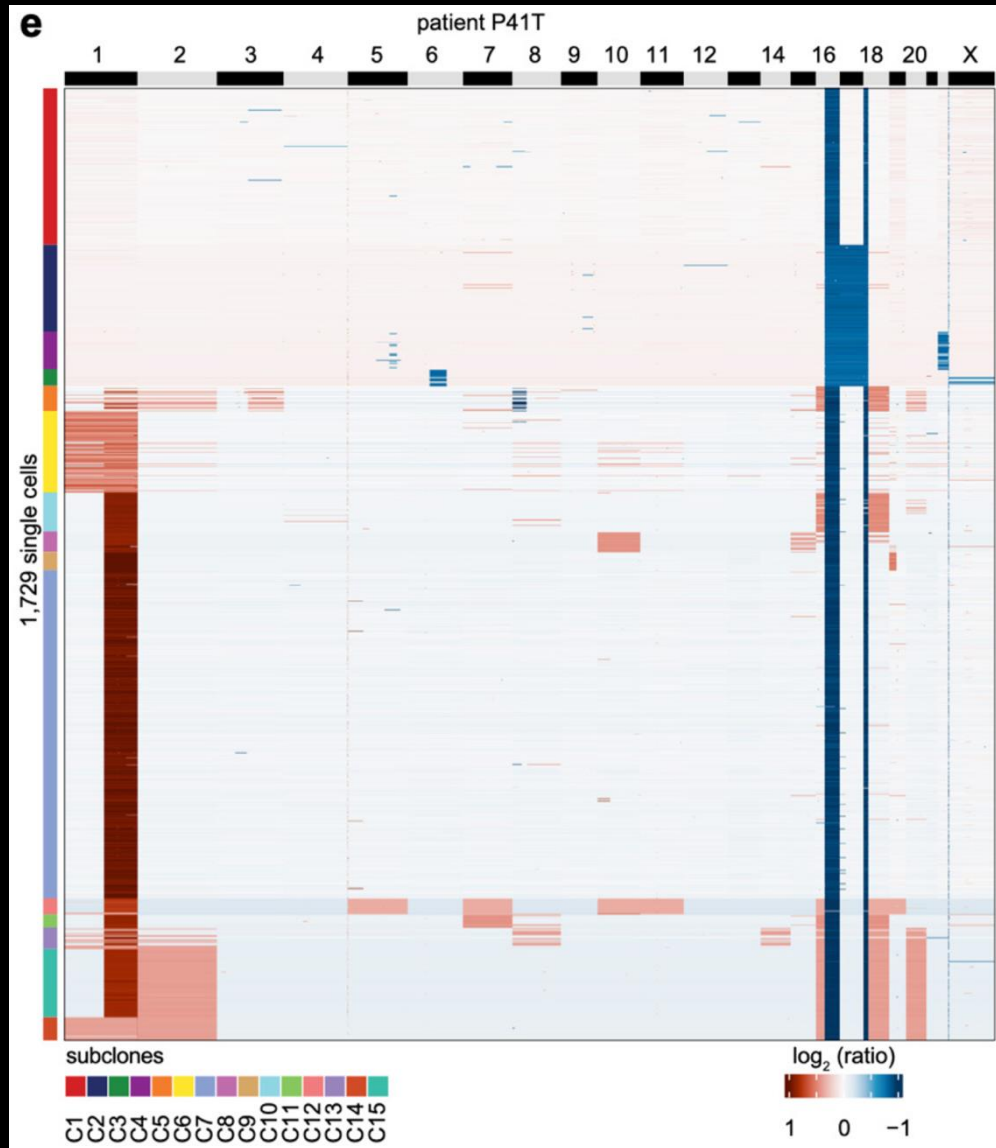
1496 samples



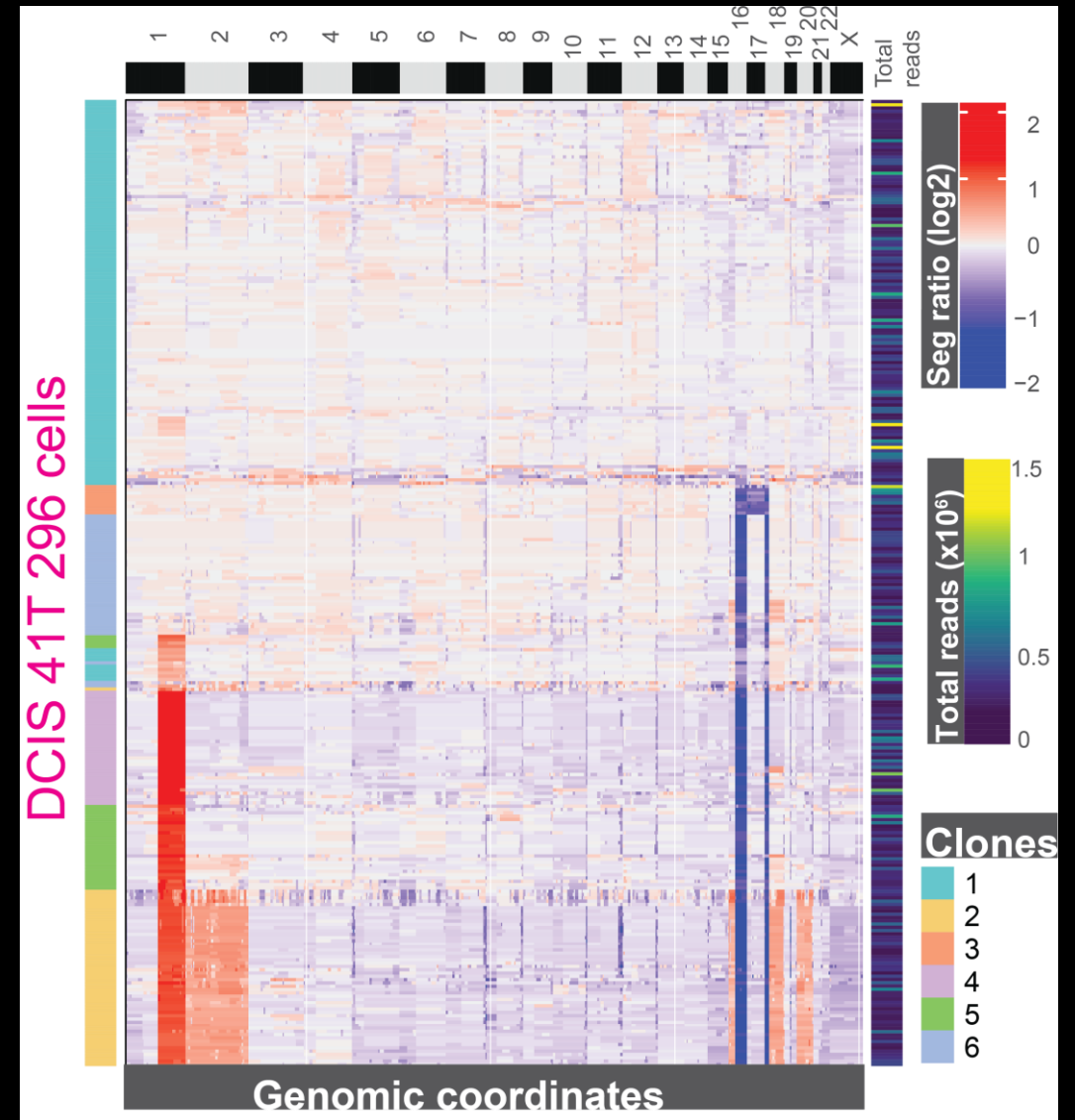
Lin, Yiyun, et al. "Normal breast tissues harbor rare populations of aneuploid epithelial cells." *Cancer Research* 84.6_Supplement (2024): 780-780.

Lineage tracing through CNVs

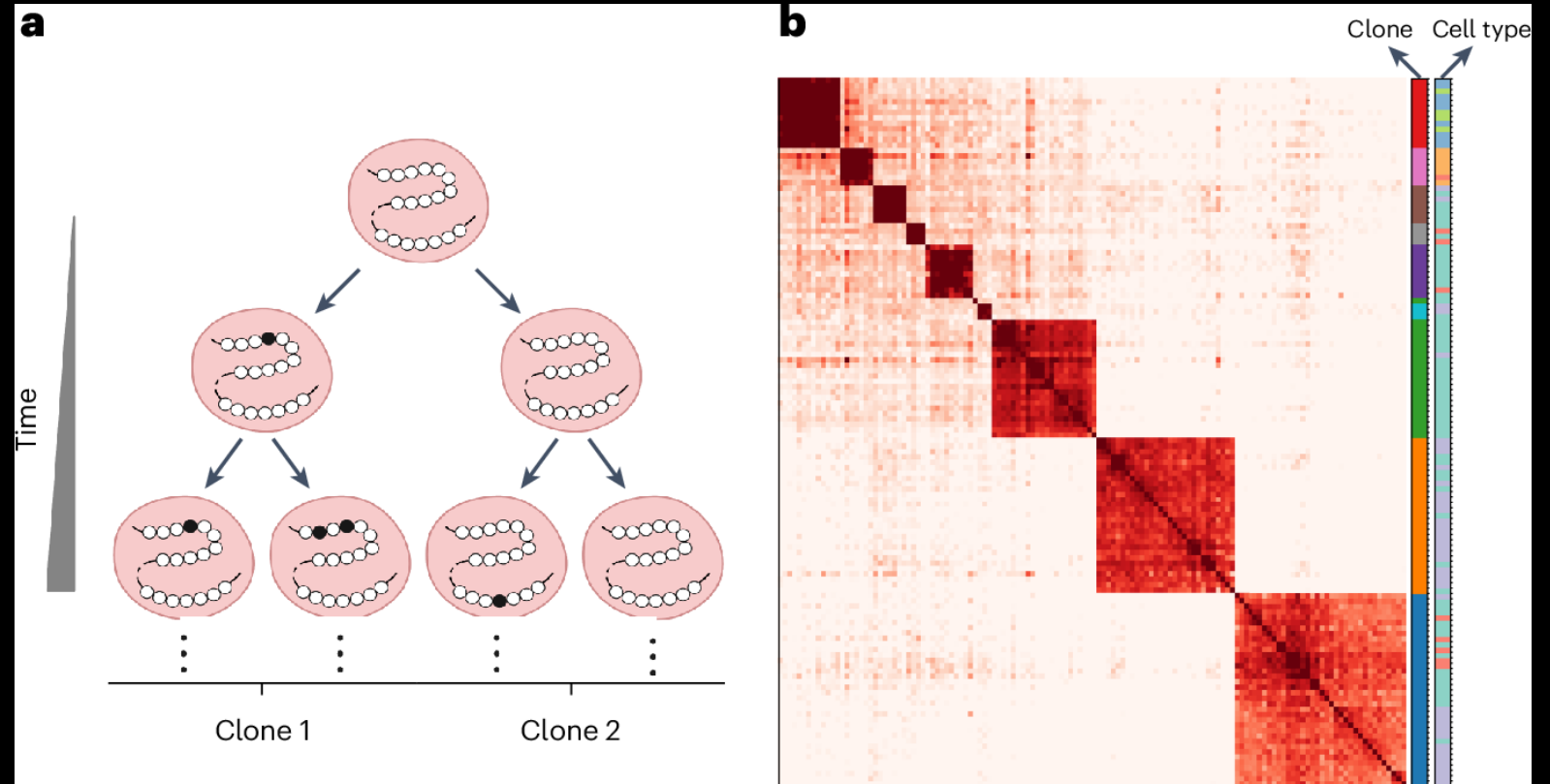
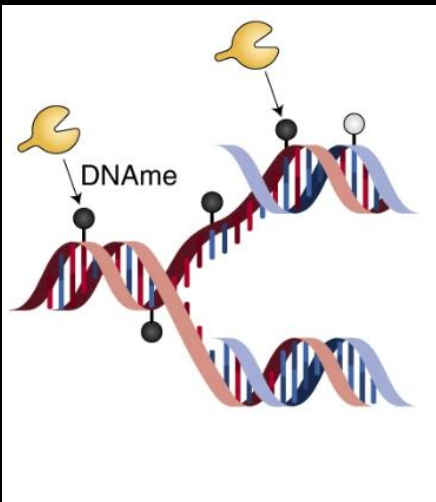
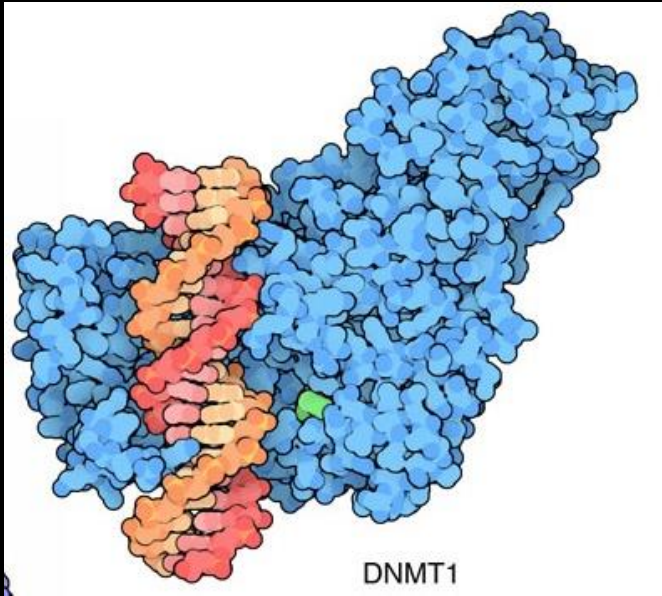
scWGS



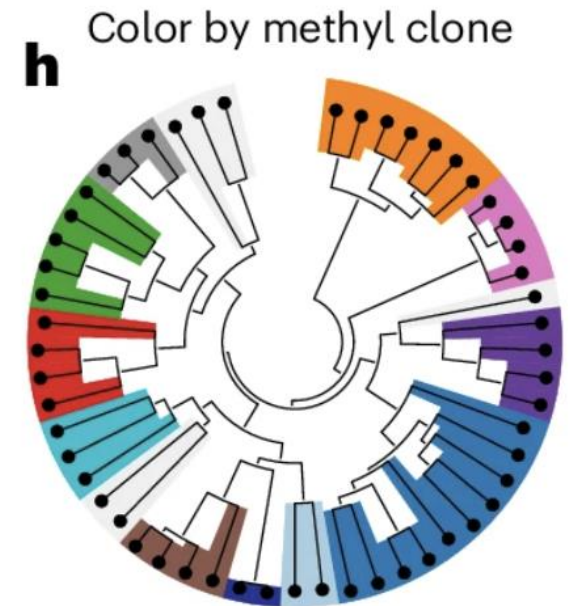
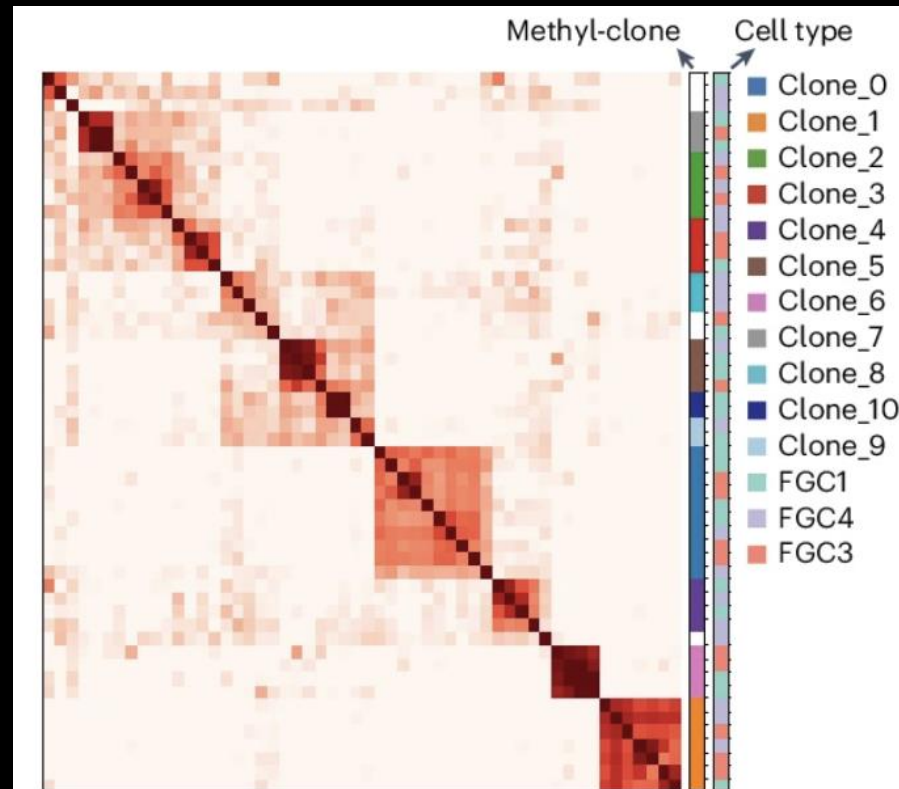
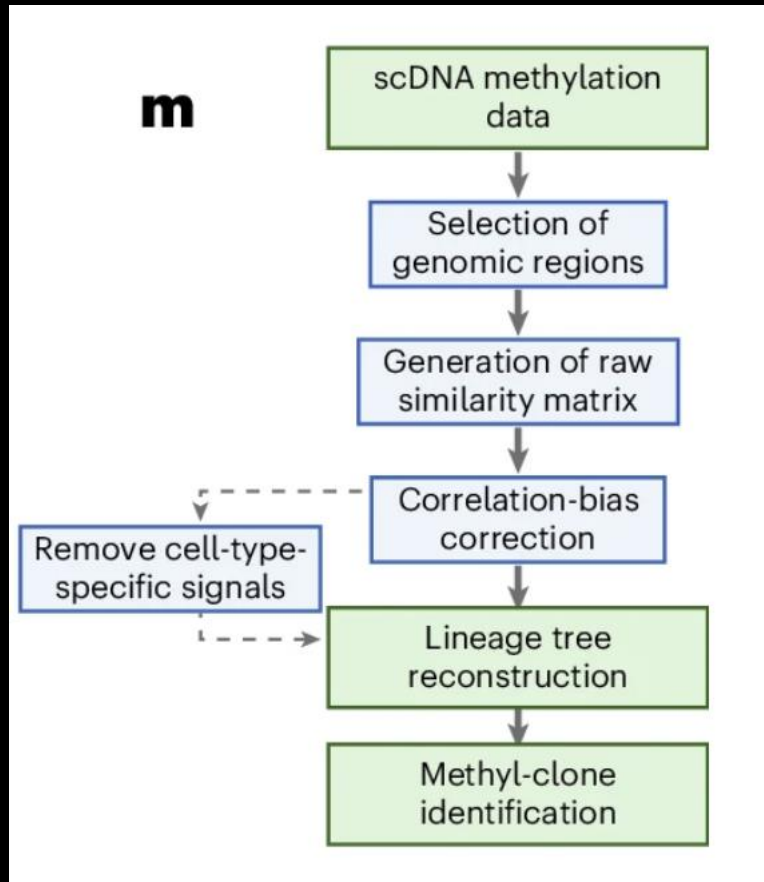
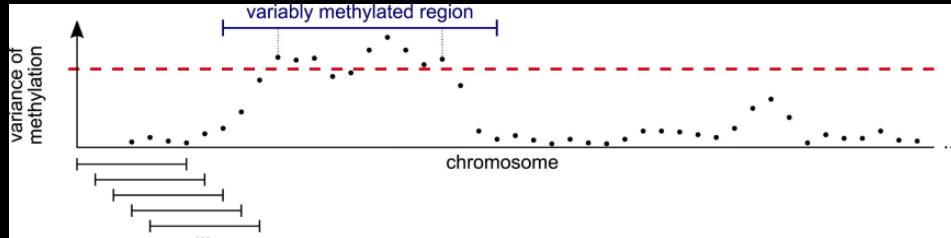
scMET



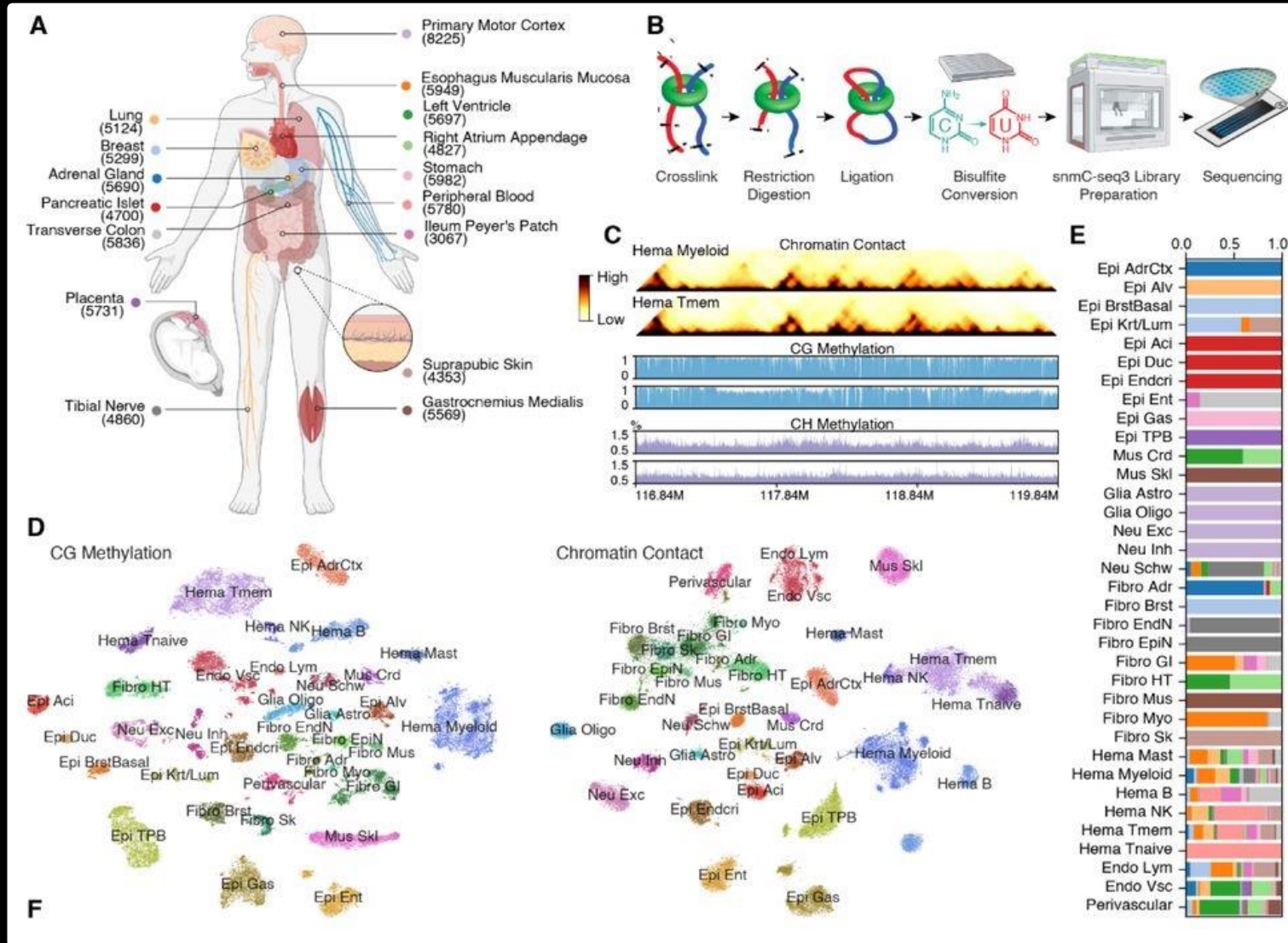
Lineage tracing through epimutations



Lineage tracing through epimutations



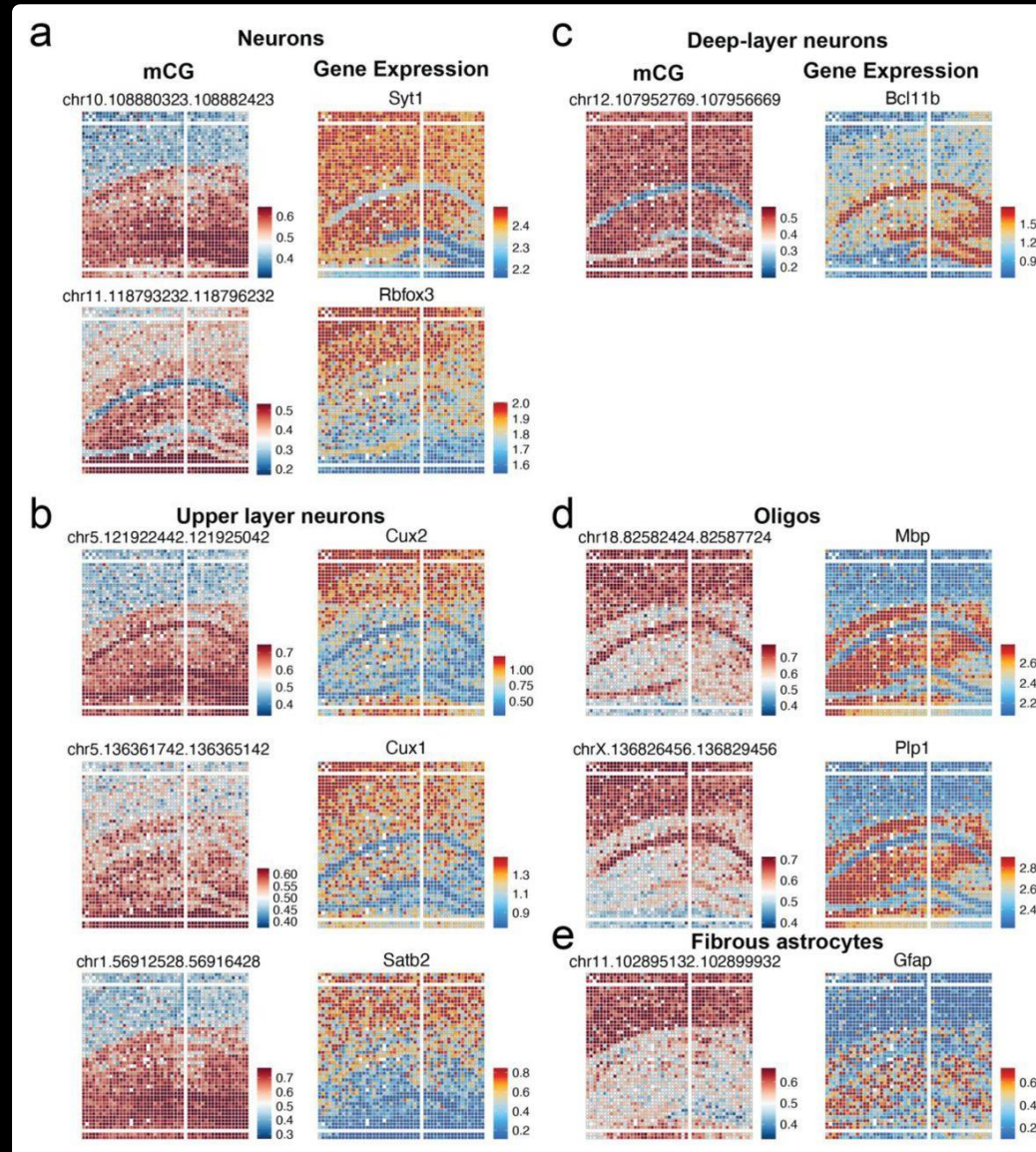
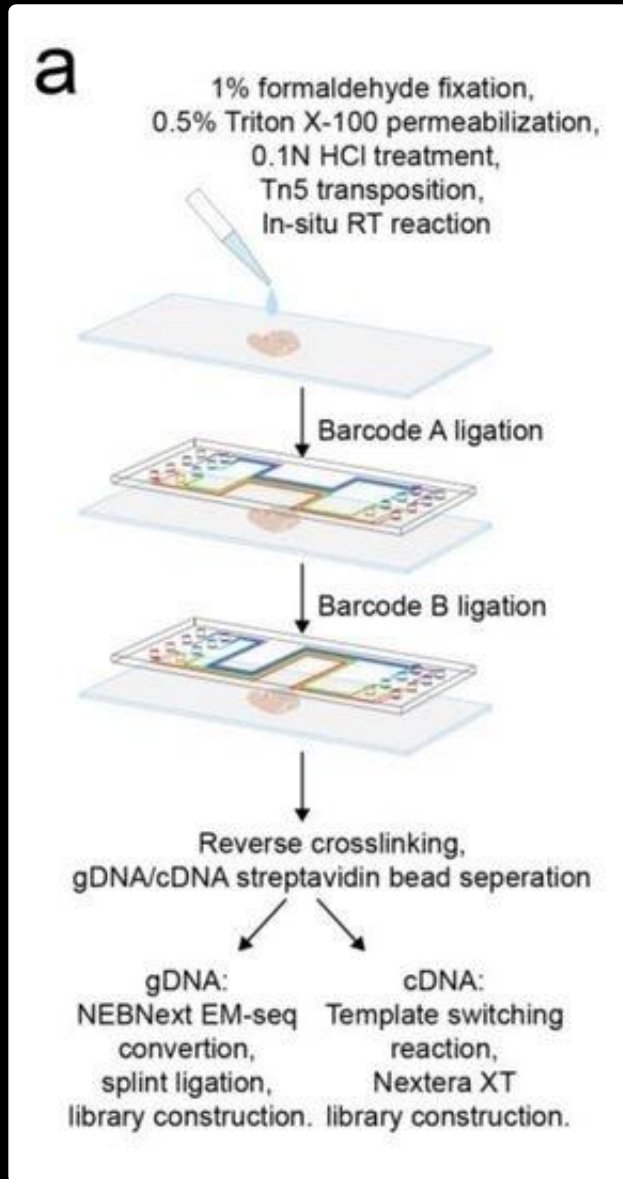
Exciting advancements in single cell methylation



Subject of our new
GCC Single Cell and
Spatial Consortium
Journal Club next
week!



Exciting advancements in single cell methylation



Yanxiang Deng is
giving a talk at BCM
on November 25th!

Conclusions

My goals:

1. Show the **commonalities** of most single-cell analyses and their use of counts data.
2. Show off relatively new assay and its **unique** challenges and analyses.
3. Get you excited about methylation and thinking about how it can be useful for you.

The workshop next month:

- Clustering single cell methylation data from three cell lines and identifying them through marker genes.
- You'll need a google account: we'll run it through [google colab](#)